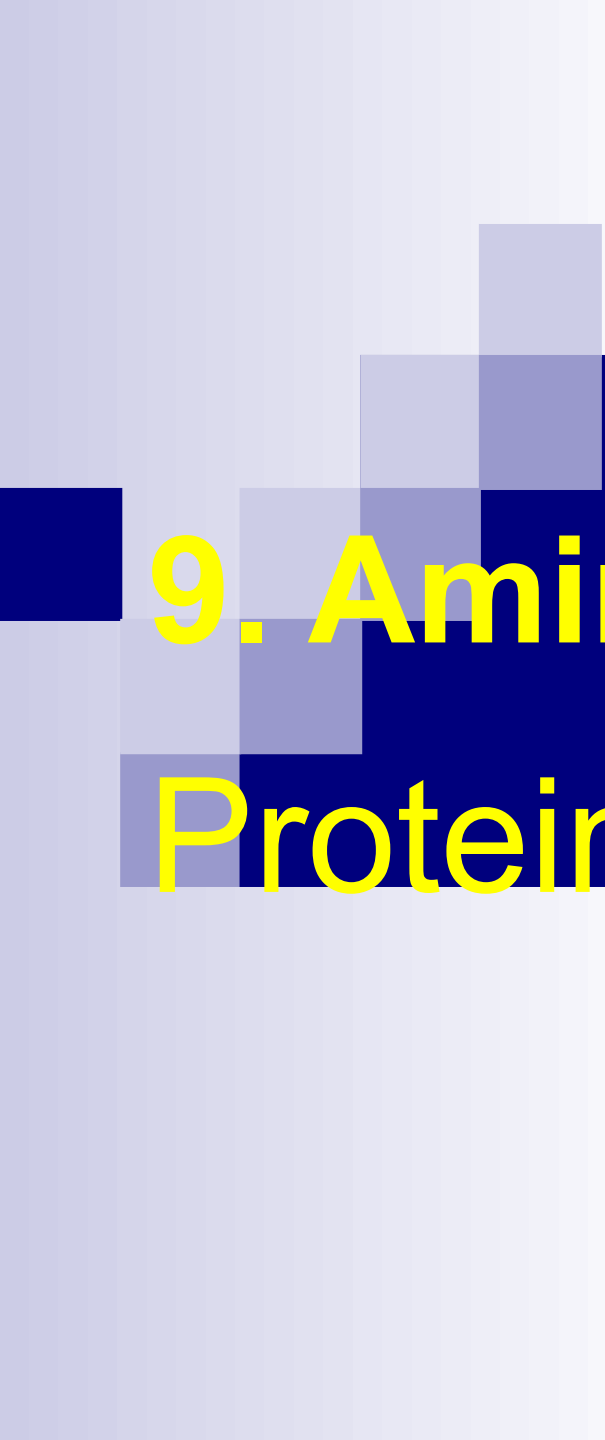
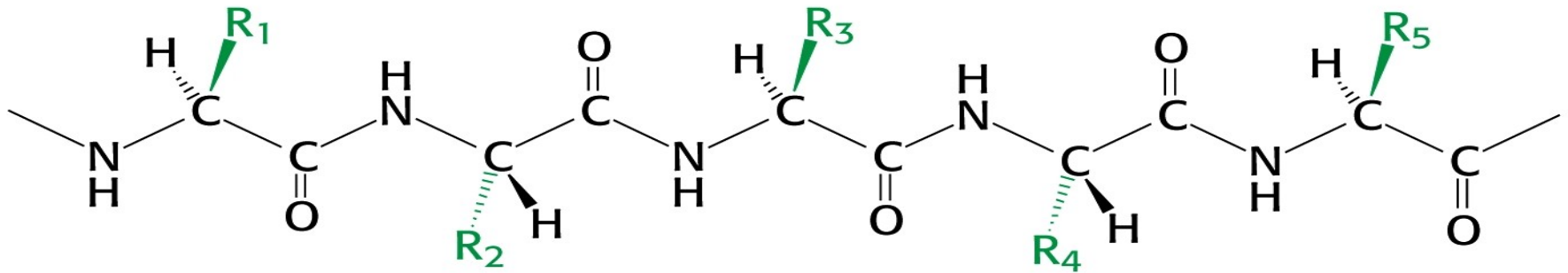


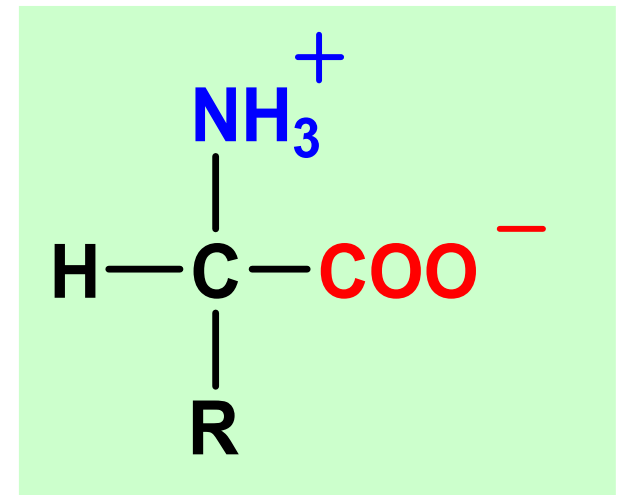


9. Amino Acid and Protein Metabolism

Reviews:



metabolism { **synthesis**
catabolism





Major content

- ◆ **Digestion and absorption** of protein
- ◆ **Normal metabolism** of amino acids
- ◆ **Special products** of amino acids



Nutritional Function of Protein

Proteins play a major role in ensuring your health well being. There are innumerable functions of proteins in the body.

- building and repairing of body tissues.

- protein makes up nearly 17 percent of the total body weight.
- For example: muscle contains about $\frac{1}{3}$ protein, bone about $\frac{1}{5}$ part and skin consists of $\frac{1}{10}$ portion. The rest part of proteins is in the other body tissues and fluids.

- Take part in some kinds of important physiological activities

- regulation of body processes and formation of enzymes and hormones, antibody.
- There are distinctive kinds of proteins, each performing a unique function in the body.

- Oxidation and supply energy

How to assess the condition of protein metabolism?

1. Nitrogen balance

the balance between the amount of nitrogen taken in (foods or the body) and the amount given off (lost or excreted)

Significance:

Measuring the amount of intake and losses of total nitrogen can help us to know the general situation of protein metabolism.

nitrogen balance

★ **positive:** synthesis $>$ degradation

(e.g., growth, body building)

★ **negative:** synthesis $<$ degradation

(e.g., starvation, trauma, cancer cachexia)

★ **Equilibrium:** synthesis = degradation

(healthy adults eating a balanced diet)

2. Physical requirements of proteins

- Lowest requirement:
30~50g/day
- Recommend requirement:
80g/day (65kg man)

Amino acids are not stored by the body, must be obtained from the diet, synthesized de novo.

Some sources of dietary protein include:

Meat, poultry and fish

Eggs, Dairy products

Seeds and nuts

Beans and lentils

Soy products

Grains, especially wheat and rice, barley and corn.



3. Nutrition value of proteins

(A) Essential amino acids :

some Amino acids that cannot be synthesized by the body and must be obtained from the diet.

Eight amino acids are generally regarded as essential for humans:

phenylalanine, valine, threonine, tryptophan, isoleucine, methionine, leucine, and lysine

(B) Non- essential amino acids

other 12 kinds of AAs, the non-essential or dispensable amino acids can be synthesized in the body either other roadways.

Note:

- a **Arg** is synthesized in the urea cycle, but the rate is too slow to meet the needs of growth in children
- b **Met** is required to produce cysteine if the latter is not supplied adequately by the diet.
- c **Phe** is needed in larger amounts to form Tyr if the latter is not supplied by the diet.

His and **Arg** are essential AAs for infants and children.

(4) nutrition value

A protein's nutritional value is judged by how many of the essential amino acids it provides and in what quantity.

Different foods contain different numbers and amounts of the essential amino acids



lysine



tryptophan



(5) Complementary effect of dietary proteins

- Two or more plant proteins are consumed together which complement each other in essential amino acid content.



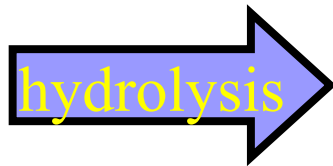


Digestion and Absorption of protein

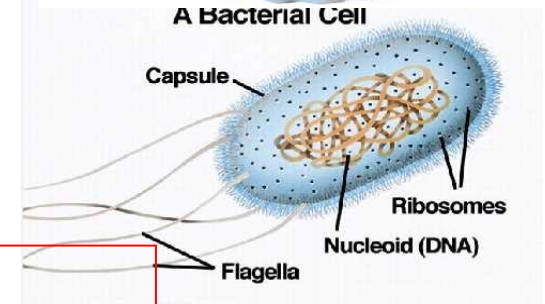
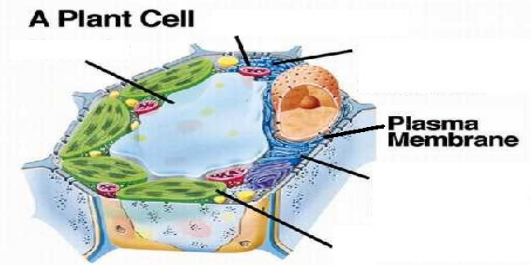
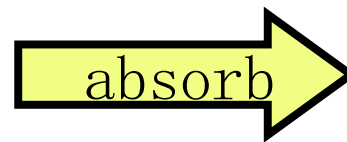
2.1 Digestion



Dietary protein



**Amino
acids**



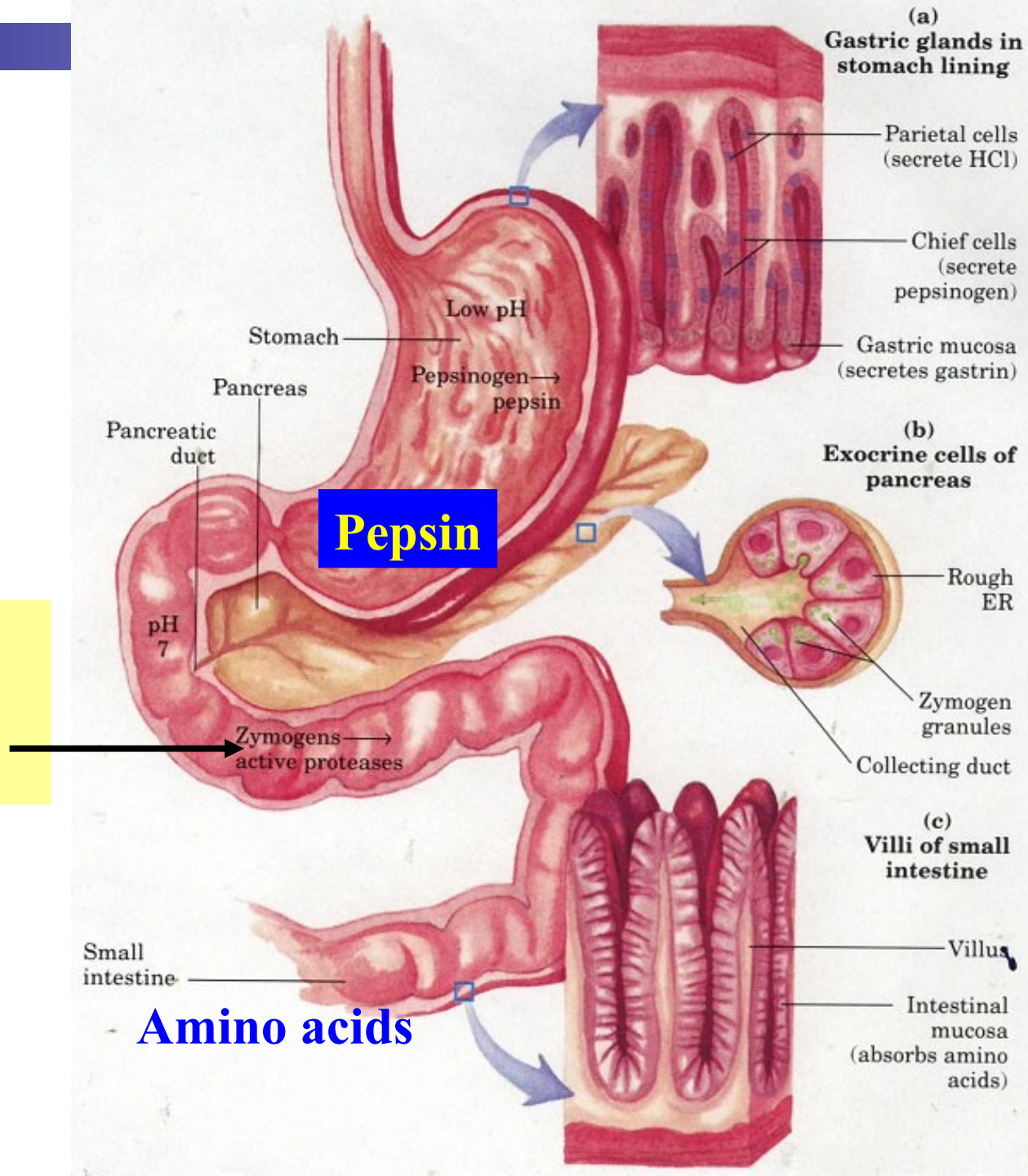
Significance:

- ♦ Large  small Help to absorb
- ♦ eliminate the species specificity and antigenicity, avoid allergy , toxic reaction.

site:
stomach,
small intestine

Chymotrypsin,
trypsin,
and exopeptidases

Proteolytic enzymes
of pancreatic juice



■ Initiated in stomach

◆ enzymes: **pepsin**



HCl from parietal cells

Pepsin

■ Stomach pH 1.6 to 3.2

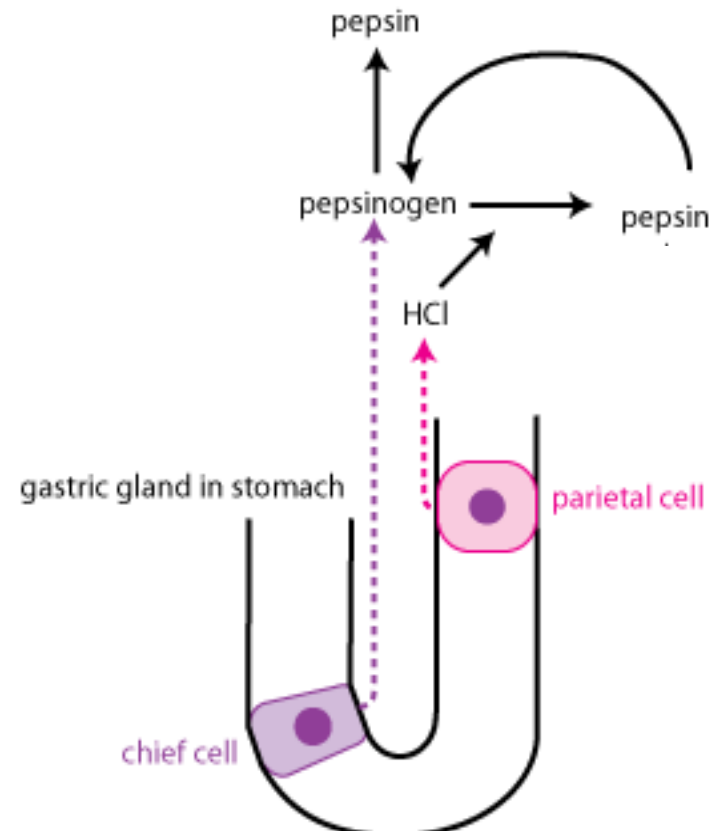
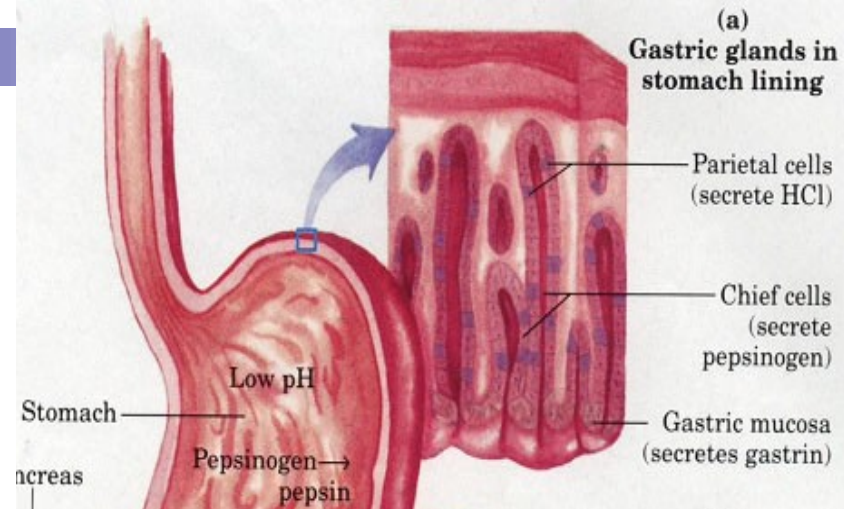
■ Pepsinogen from chief cells

◆ The substrate mainly are
phenylalanine, tyrosine, tryptophan

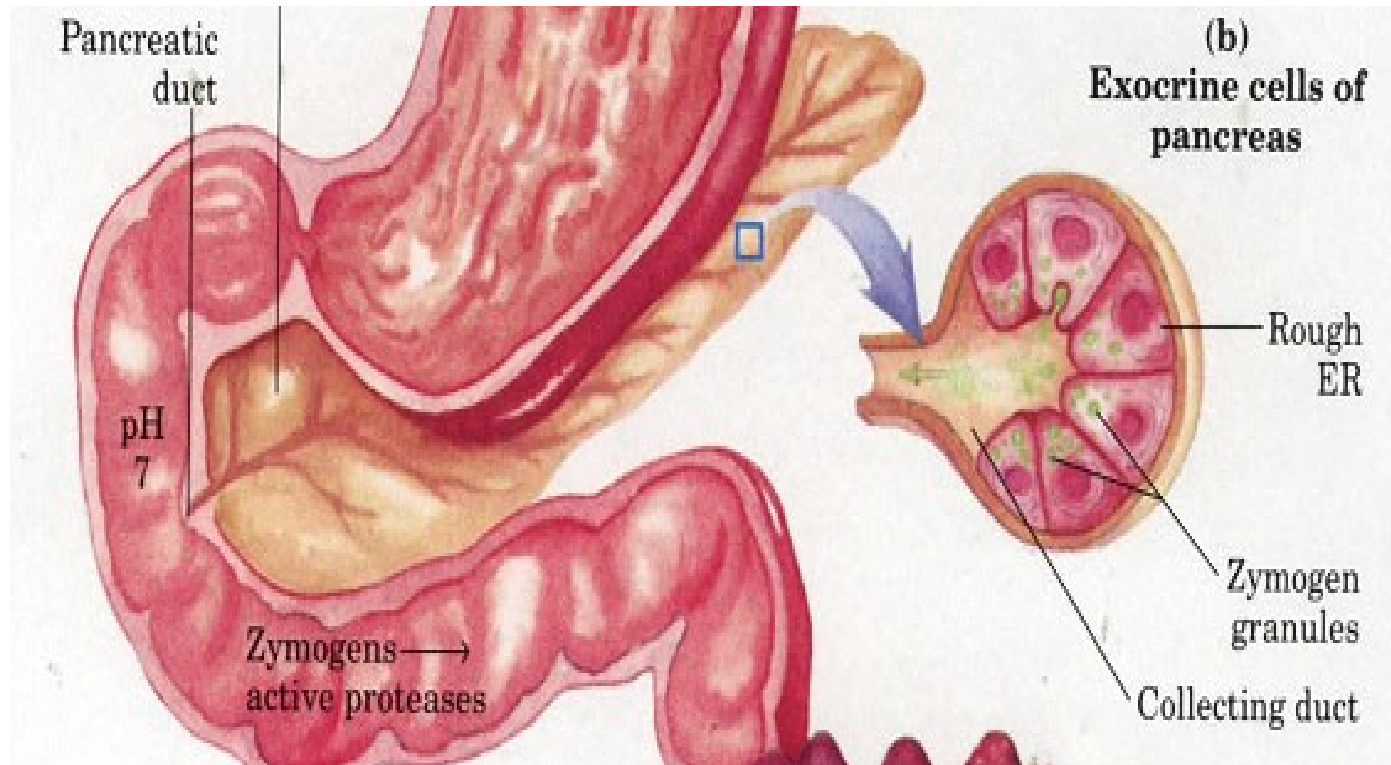
Aromatic amino acids

◆ Products:

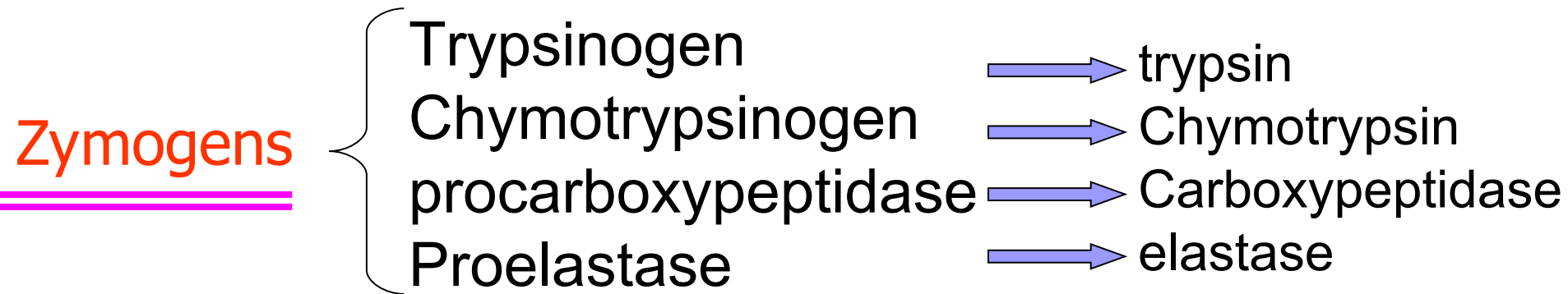
insoluble protein, soluble protein,
polypeptides and amino acids



Protein Digestion – Small Intestine



■ Pancreatic enzymes secreted



Zymogens

A zymogen is the **inactive precursor** of an enzyme.

Activation of zymogen

A inactive zymogen become active enzyme.

In a zymogen, a peptide blocks the active site of the enzyme. Cleaving off this peptide activates the enzyme.



Significance:

1. avoids self-digestion:

This is necessary to prevent the digestive enzymes from autodigesting the cells that produce them.

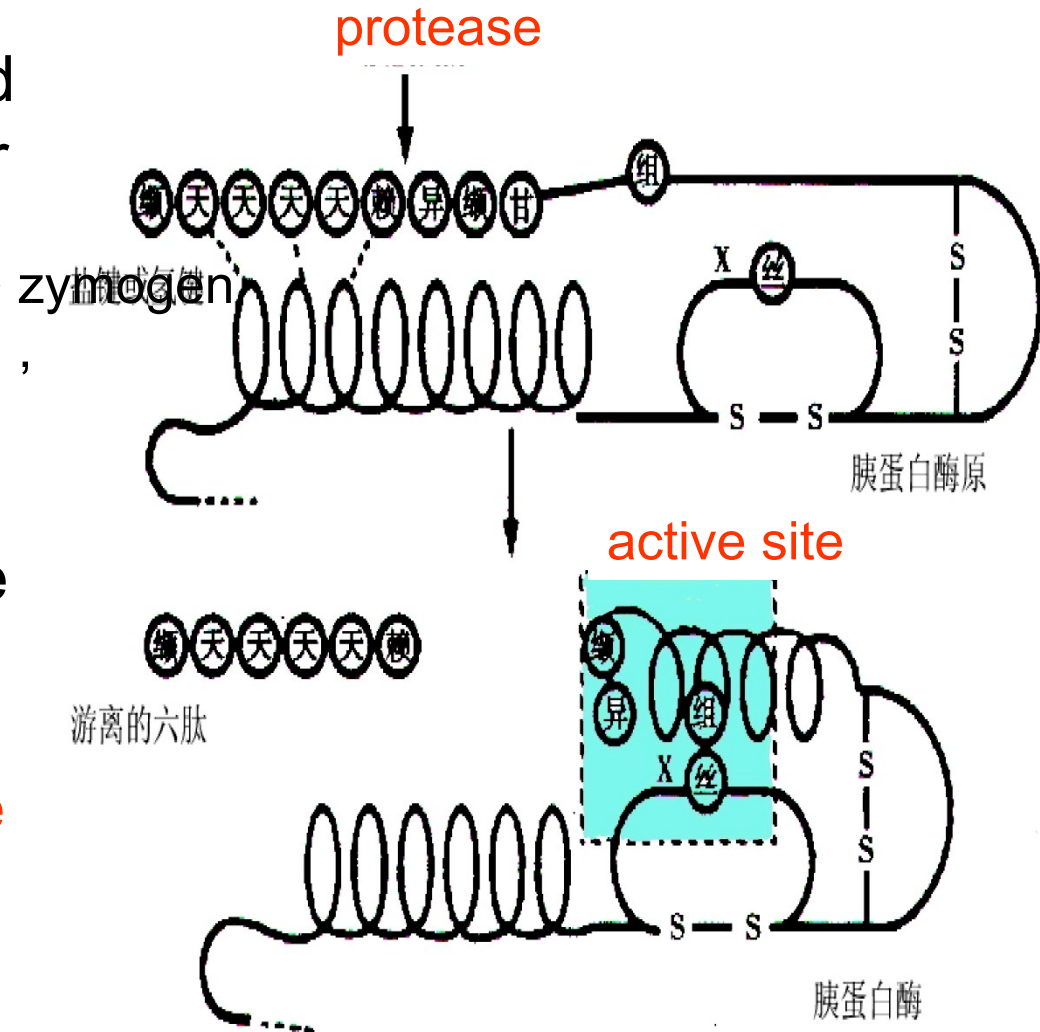
2. stored and transported safely :

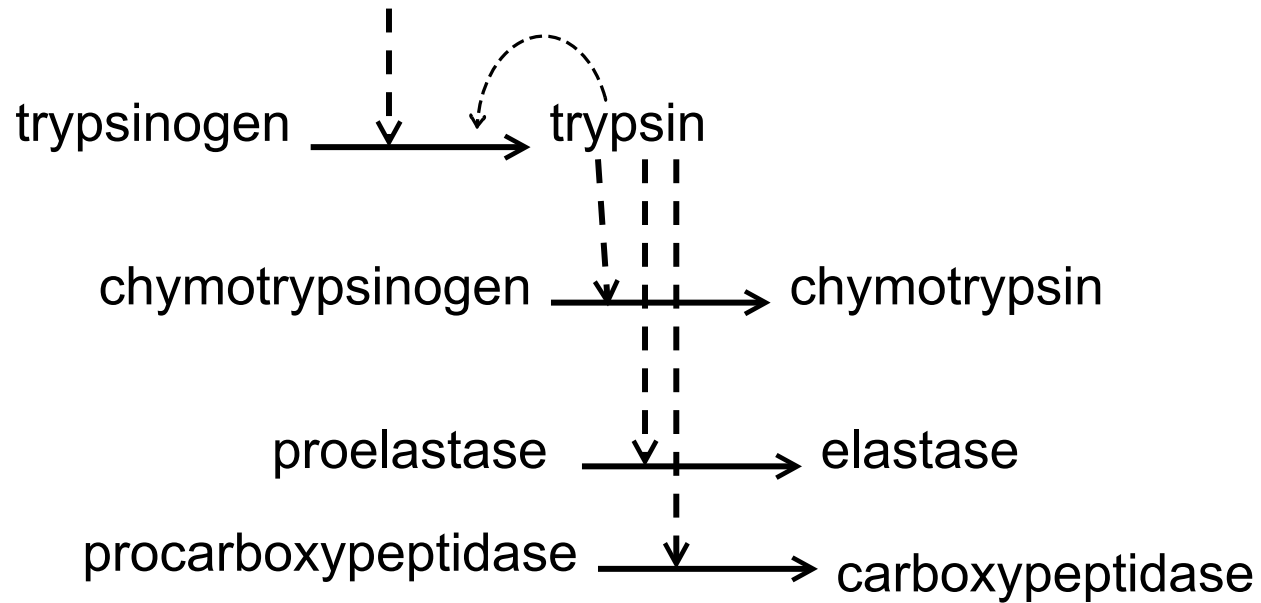
The body typically secretes zymogens rather than active enzymes because they can **be stored and transported safely** without harm to surrounding tissues, and released when conditions are favorable for optimal activity.

In a zymogen, a peptide blocks the active site of the enzyme. Cleaving off this peptide activates the enzyme.

The molecule is composed of amino acids strung together into a peptide. When the zymogen is in the presence of **protease**, some of the amino acids are removed.

This cleavage renders the zymogen a functional enzyme by changing the shape of the peptide and **forming the active site** where enzymatic action will occur.





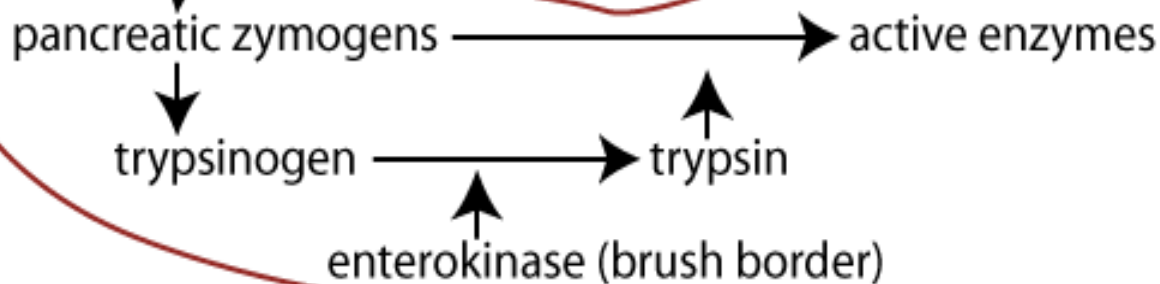
duodenum

common bile duct

pancreatic duct

cascade reaction

Amplification effect



Protein Digestion – Small Intestine

Proteolytic enzymes of pancreatic juice

endopeptidases { **trypsin:** Arg, Lys (C)
chymotrypsin: Tyr, Trp,
Phe, Met, Leu (C)
elastase: Ala, Gly, Ser (C)

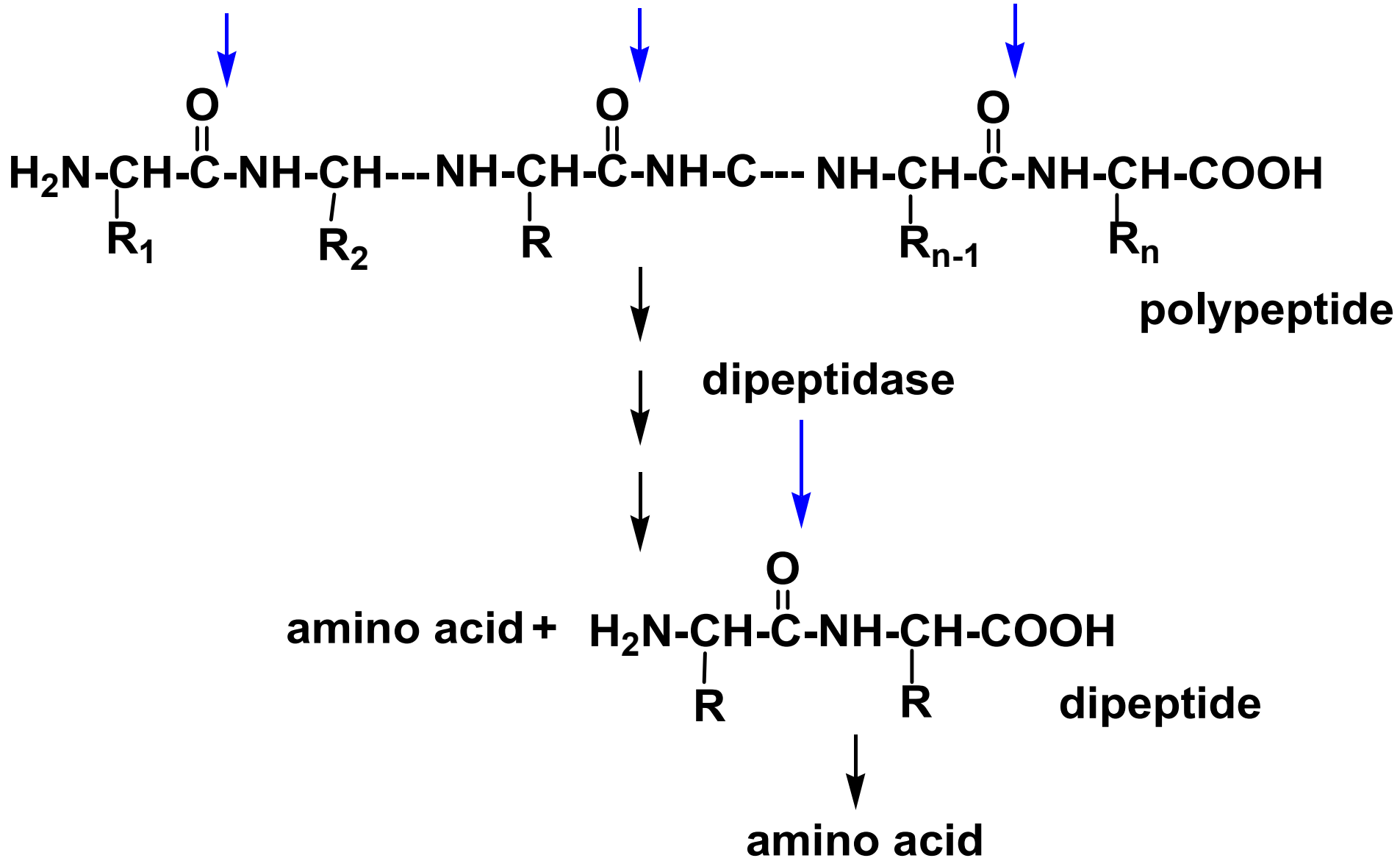
exopeptidases { carboxypeptidase
aminopeptidase



amino peptidase

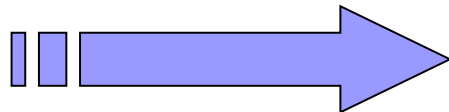
endopeptidase

carboxy peptidase



Protein Digestion

- Proteins are broken down to
 - Tripeptides
 - Dipeptides
 - Free amino acids

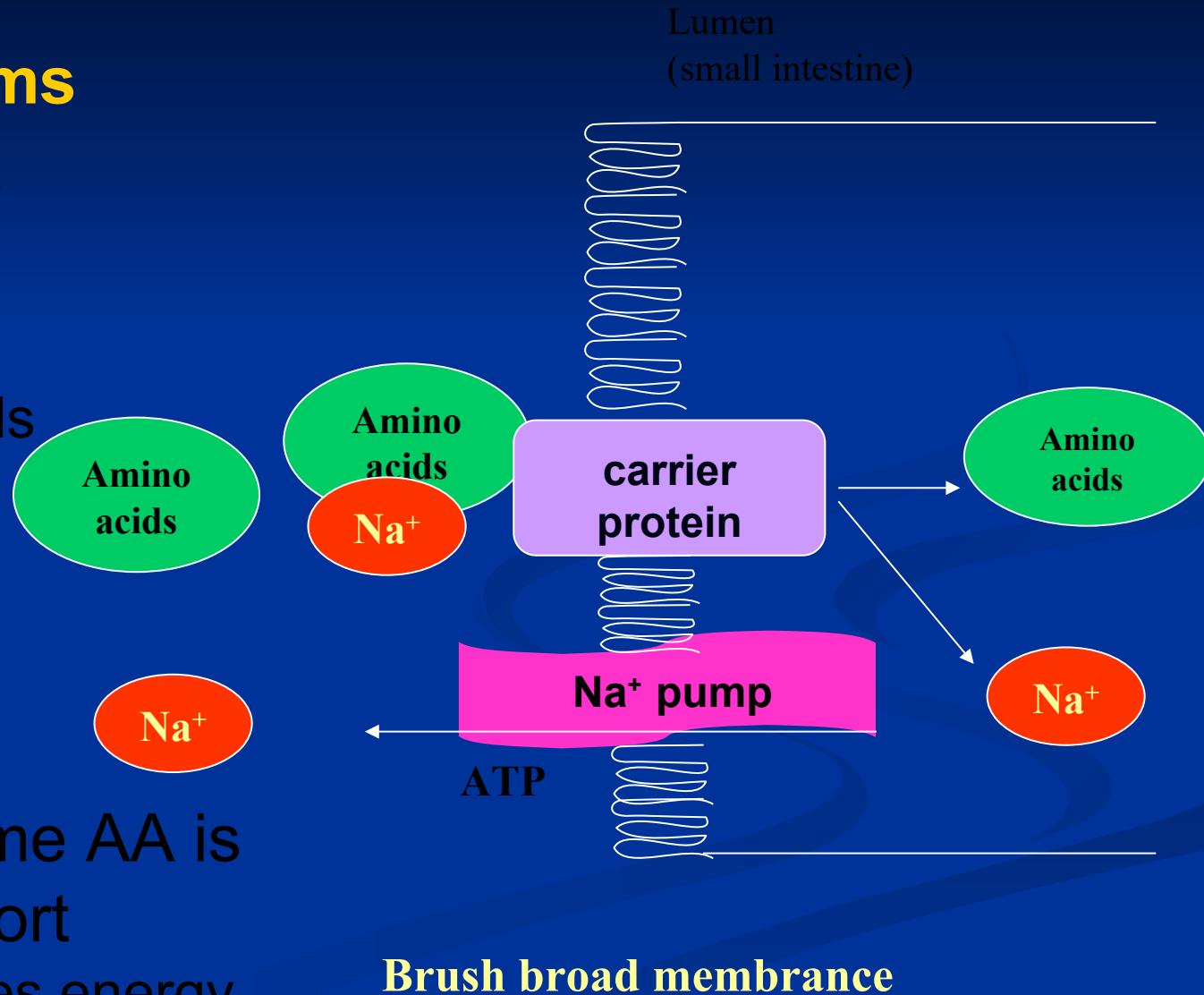
 **2.2 absorption**

Free Amino Acid Absorption

■ Carrier systems

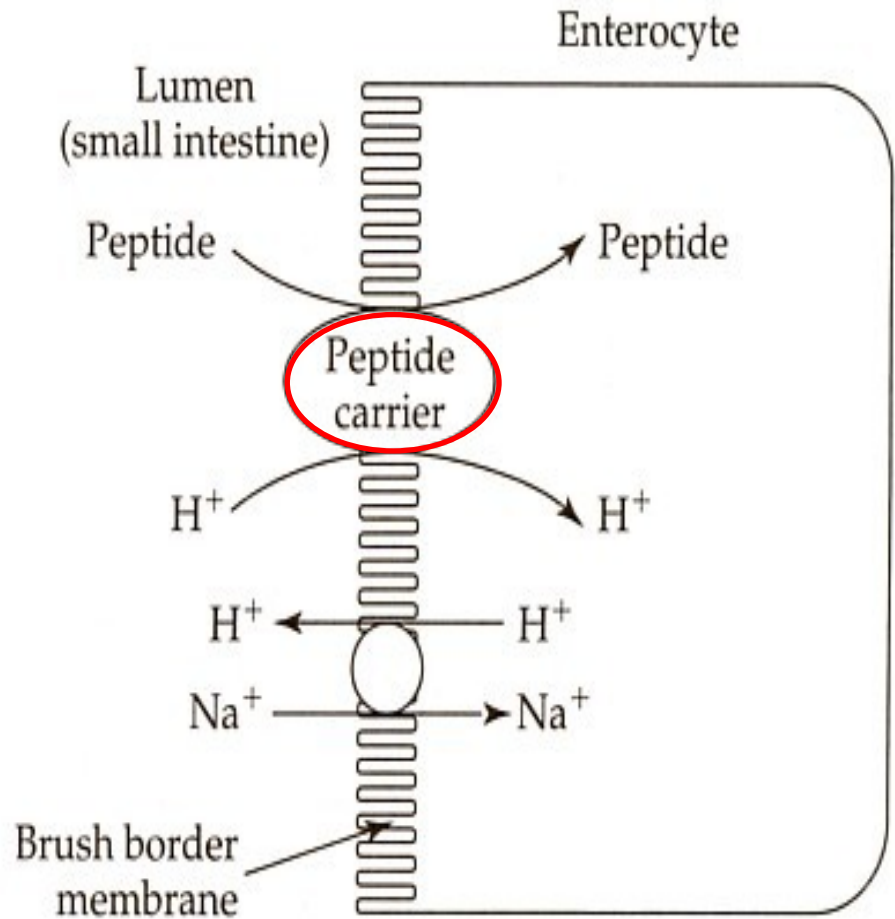
- Neutral AA
- Basic AA
- Acidic AA
- Amino acids

- Entrance of some AA is via active transport
Requires energy



Peptide Absorption

- Form in which the majority of protein is absorbed
- More rapid than absorption of free amino acids
- Active transport
 - Energy required
- Metabolized into free amino acids in enterocyte
- Only free amino acids absorbed into blood





General Metabolism of Amino Acid

§ 3.1 Protein turnover

the balance between **protein synthesis** and **protein degradation** .

- In healthy adults, the total amount of protein in the body remains constant, because **the rate of protein synthesis** is just sufficient to replace the **protein that is degraded**. this process is called protein turnover.
- **Rapid protein turnover** ensures that some regulatory proteins are degraded so that the cell can respond to constantly changing conditions.

half-life

Half-life is the period of time it takes for a substance undergoing decay to decrease by half.

Protein	turnover rate (halflife)
enzymes	7-10 minutes
in liver	10 days
in plasma	10 days
hemoglobin	120 days
muscle	180 days
collagen	1000 days

Examples of protein turnover in the body

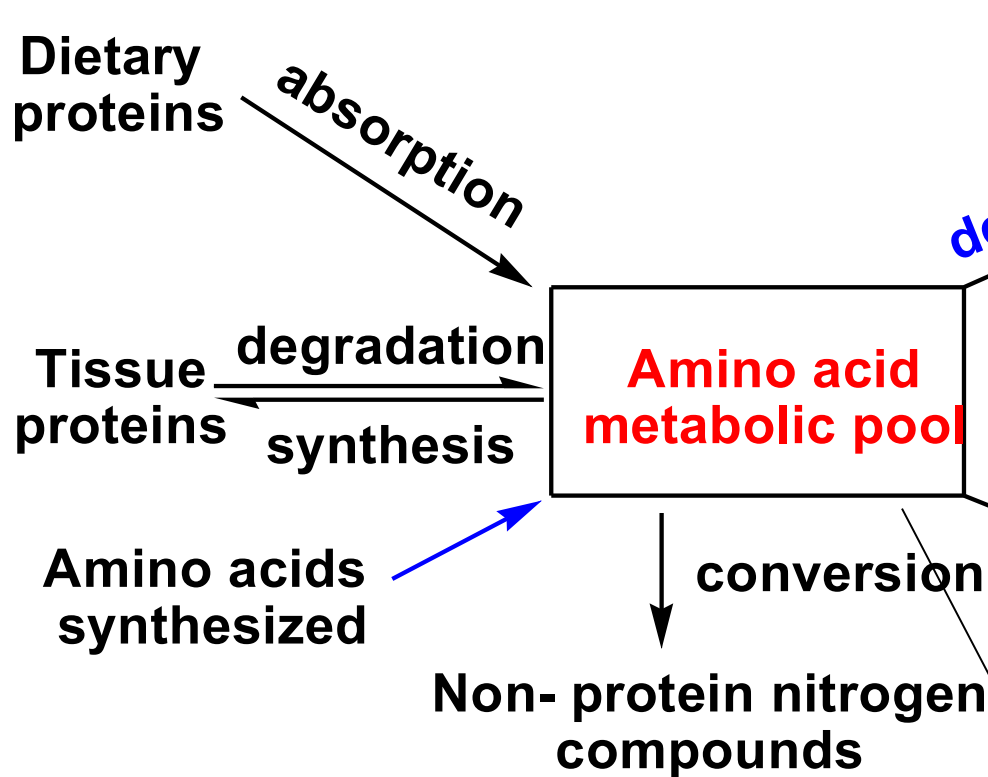
Amino acid pool:

amino acids in intracellular and extracellular fluids.

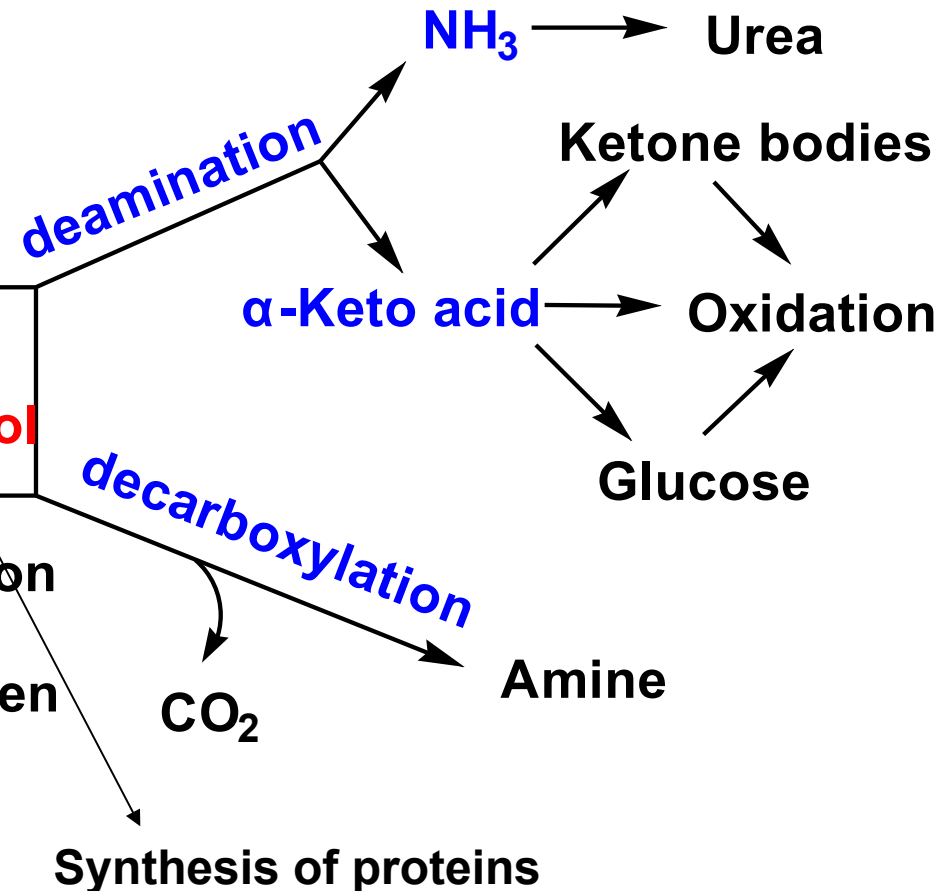
	Amino acids%
muscle	50%
liver	10%
kidney	4%
blood	1~6%

§ 3.1 The sources and fates of AAs

Sources of amino acids



Fates of amino acids





§ 3.3 The catabolism of AAs

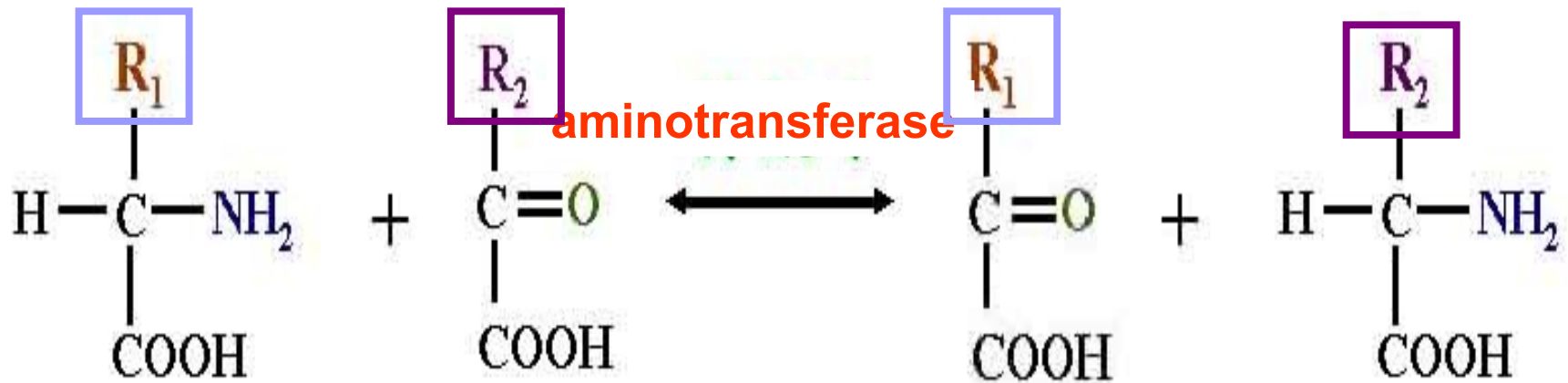
1. Deamination of AAs

Two types:

transamination

oxidative deamination

(1) Transamination



Transamination is the process by which an amino group, **usually from glutamate**, is transferred to an α -keto acid, with formation of the corresponding amino acid plus α -ketoglutarate.

Key points:

- ① reversible: **Transaminases** (aminotransferases) catalyze the reversible reaction at right.
- ② Lys and Pro cannot be transaminated.
- ③ Aminotransferases utilize a coenzyme - **pyridoxal phosphate (PLP)**, - which is derived from **vitamin B₆**.



Transaminases **equilibrate amino groups** among available α -keto acids.

This permits synthesis of non-essential amino acids, using amino groups from other amino acids & carbon skeletons synthesized in a cell.

Thus a balance of different amino acids is maintained, as proteins of varied amino acid contents are synthesized.

Although the amino N of one amino acid can be used to synthesize another amino acid, **N must be obtained in the diet as amino acids** (proteins).



In addition to equilibrating amino groups among available α -keto acids, transaminases **funnel amino groups** from excess dietary amino acids to those amino acids (e.g., glutamate) that can be **deaminated**.

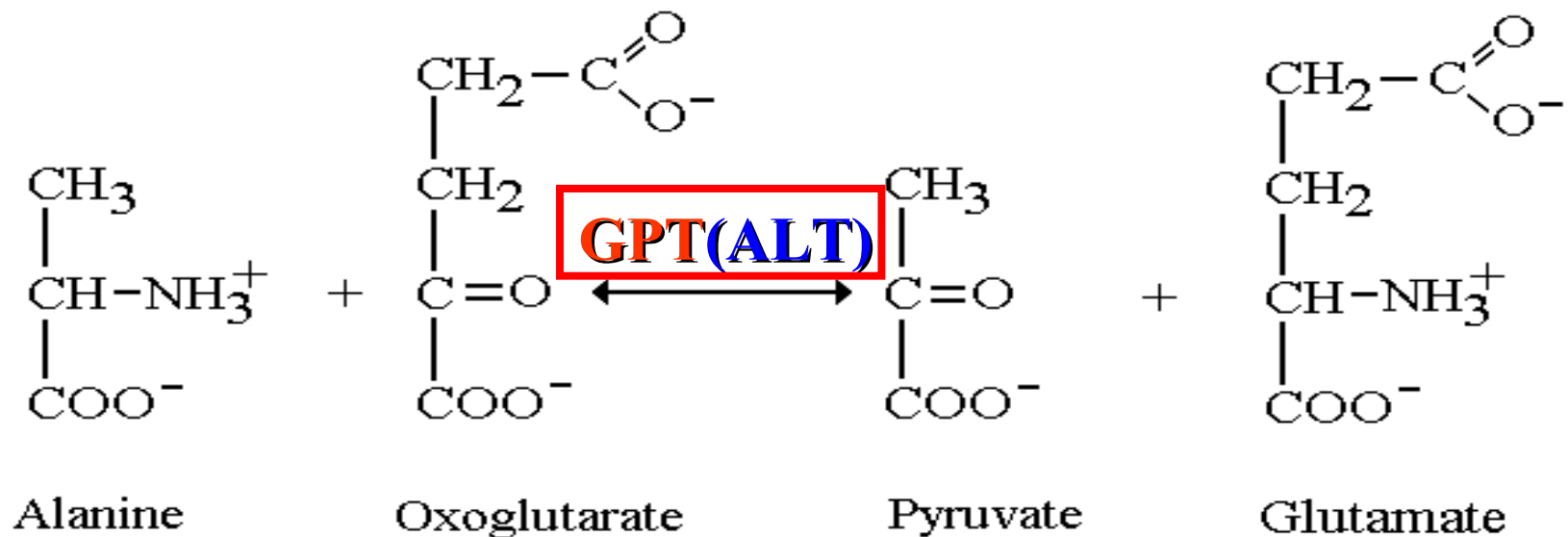
Carbon skeletons of deaminated amino acids can be catabolized for energy, or used to synthesize glucose or fatty acids for energy storage.

Only a few amino acids are deaminated directly.

Two important transaminases:

1. GPT (serum glutamate pyruvate transaminase)

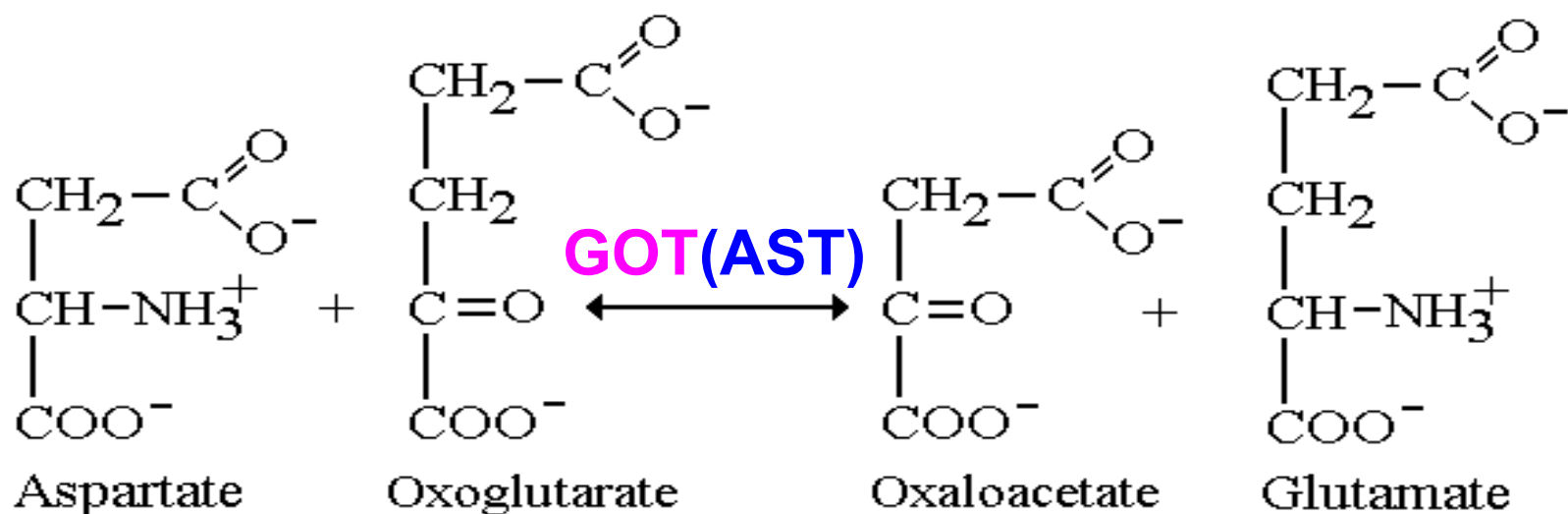
/ Alanine transaminase (**ALT**)



2. GOT

- (serum glutamate oxaloacetate transaminase)

/ Aspartate aminotransferase (AST)





organ	GOT	GPT	organ	GOT	GPT
heart	156000	7100	pancrease	28000	2000
liver	142000	44000	spleen	14000	1200
skeletal	99000	4800	lung	10000	700
kidney	91000	19000	serum	20	16

- 
- ALT is an enzyme produced in hepatocytes and is highly concentrated in the liver.

! ! Therefore, when the liver is injured, ALT is released into the bloodstream.

Elevated levels of ALT may indicate :

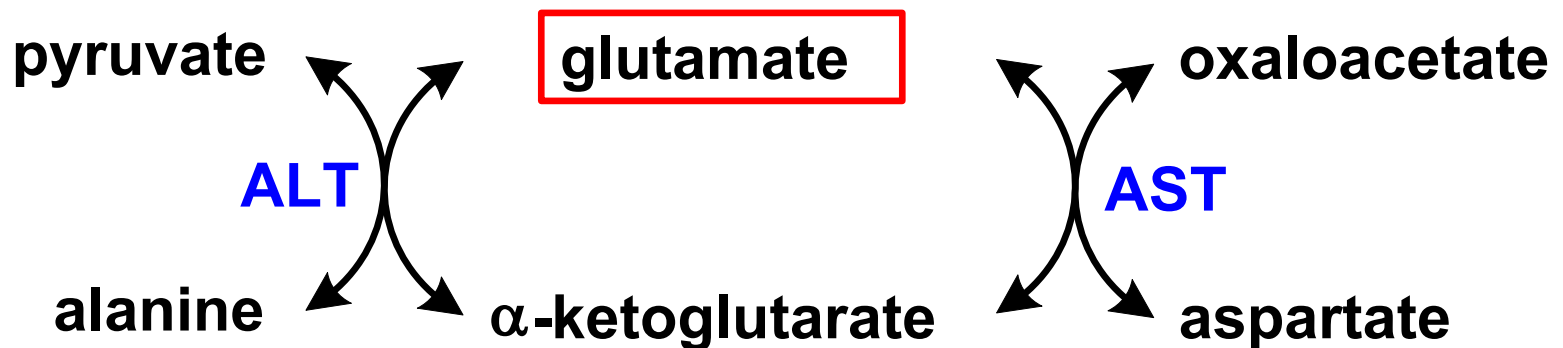
- ✓ alcoholic liver disease
- ✓ cancer of the liver
- ✓ **cholestasis or congestion of the bile ducts**
- ✓ cirrhosis or scarring of the liver with loss of function
- ✓ death of liver tissue
- ✓ **Hepatitis or inflammation of the liver**
- ✓ noncancerous tumor of the liver
- ✓ use of medicines or drugs toxic to the liver

- 
- AST also reflects damage to the hepatic cells and is less specific for liver disease. It can also be released with *heart, muscle and brain disorders.*
 - Therefore, this test may be ordered to help diagnose various heart, muscle or brain disorders, such as a *myocardial infarct* (heart attack).

Two important transaminases:

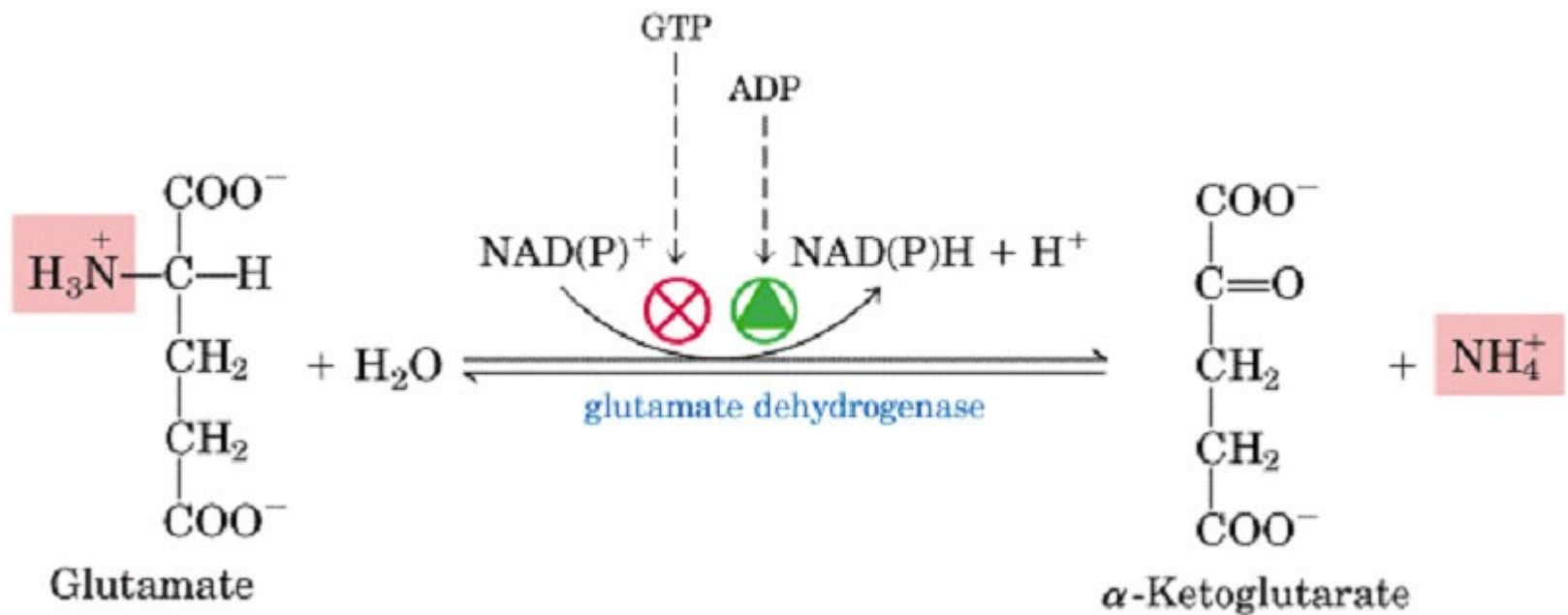
ALT: Alanine aminotransferase (in liver)

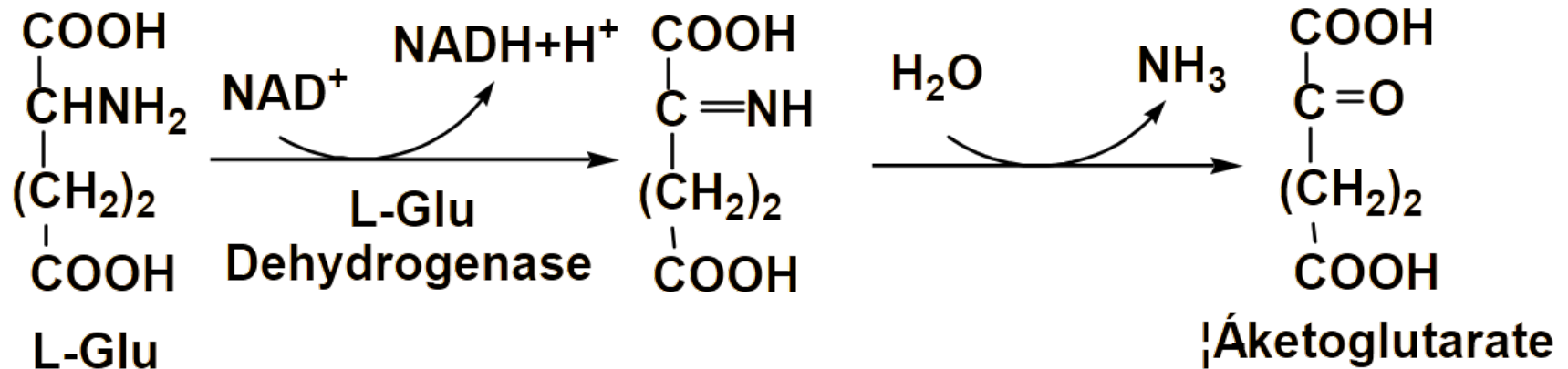
AST: Aspartate aminotransferase (in heart)



No net removal of N from the amino acid pool.

(2) Oxidative deamination

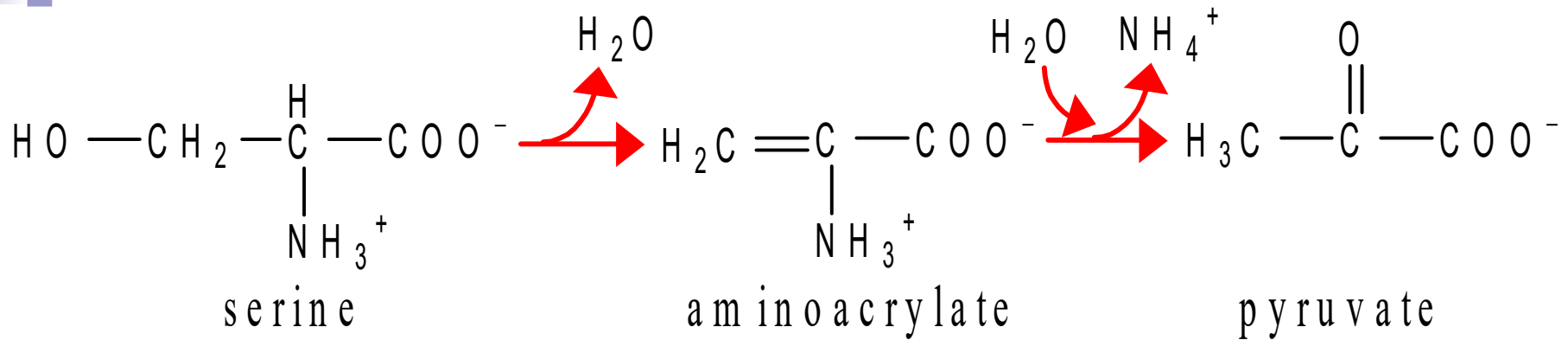




1. Glutamate Dehydrogenase catalyzes a major reaction that effects **net removal of N** from the amino acid pool.

2. It is one of the few enzymes that can use NAD^+ or NADP^+ as e^- acceptor.

Oxidation at the α -carbon is followed by hydrolysis, releasing NH_4^+ .



Serine Dehydratase

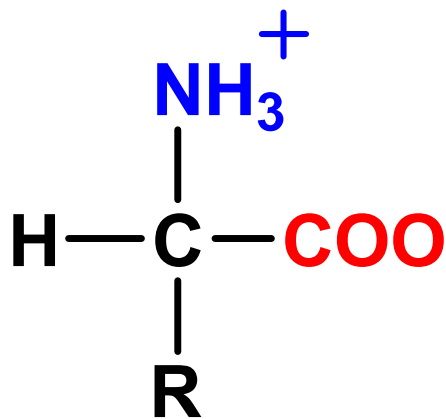
Some other pathways for deamination of amino acids:

1. **Serine Dehydratase** catalyzes:



2. **Peroxisomal** L- and D-amino acid oxidases catalyze:





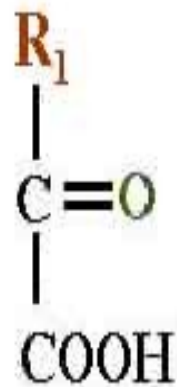
Amino acid

deamination



liver

urine



Ketone bodies

oxidation

glucose



Section 4

Metabolism of Ammonia

§ 4.1 Source and outlet of ammonia (NH₃)

1. Sources:

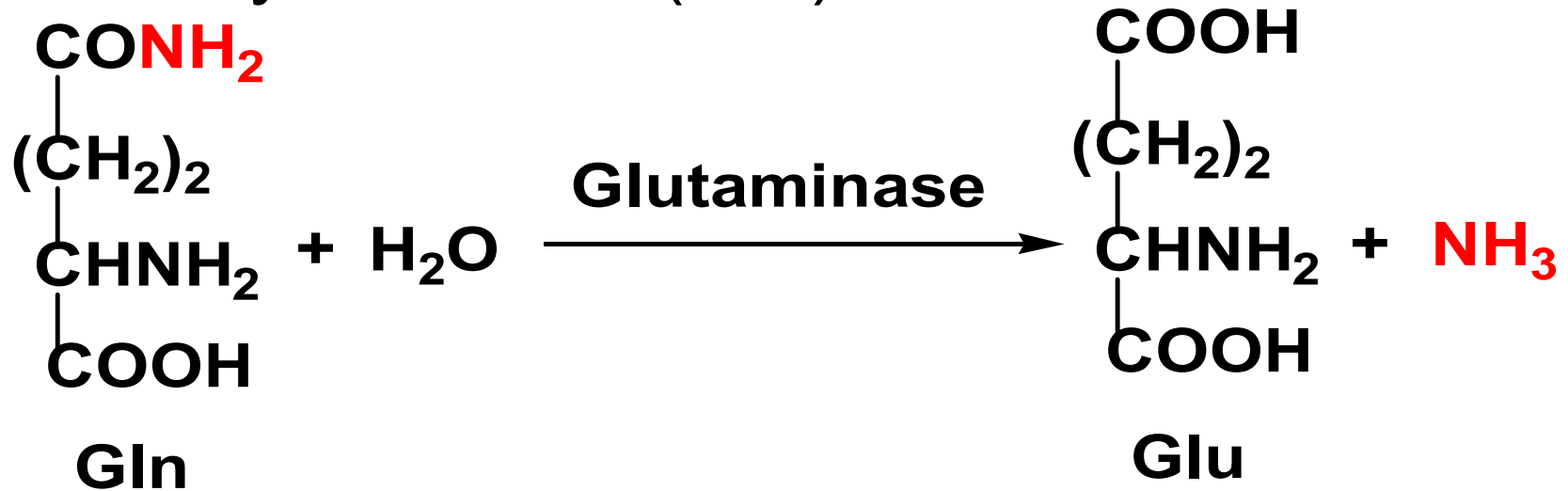
(1) Endogenous sources:

① Deamination of AAs--main source

② Catabolism of other nitrogen containing compounds.



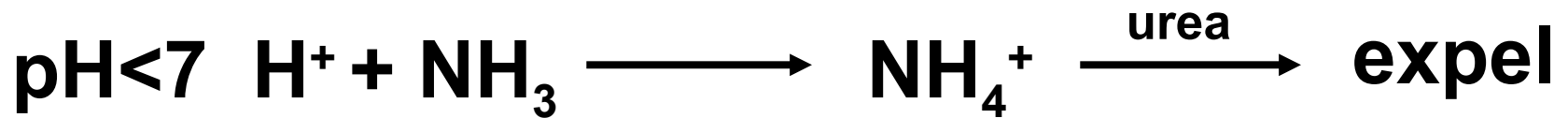
③ Kidney secretion (Gln)



(2) Exogenous sources :

- ① Putrefaction in the intestine.
- ② Degradation of urea present in fluids secreted into the GI tract

NH_3 is easy to dispersion, NH_4^+ is not .

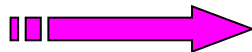


Liver desfunction

**Reduce the
absorption of
ammonia:**

acidifying diuretic

weakly acidic dialysate in colonic
dialysis



alkaline dialysate ,alkaline medication
soapsuds enema



2. Outlets:

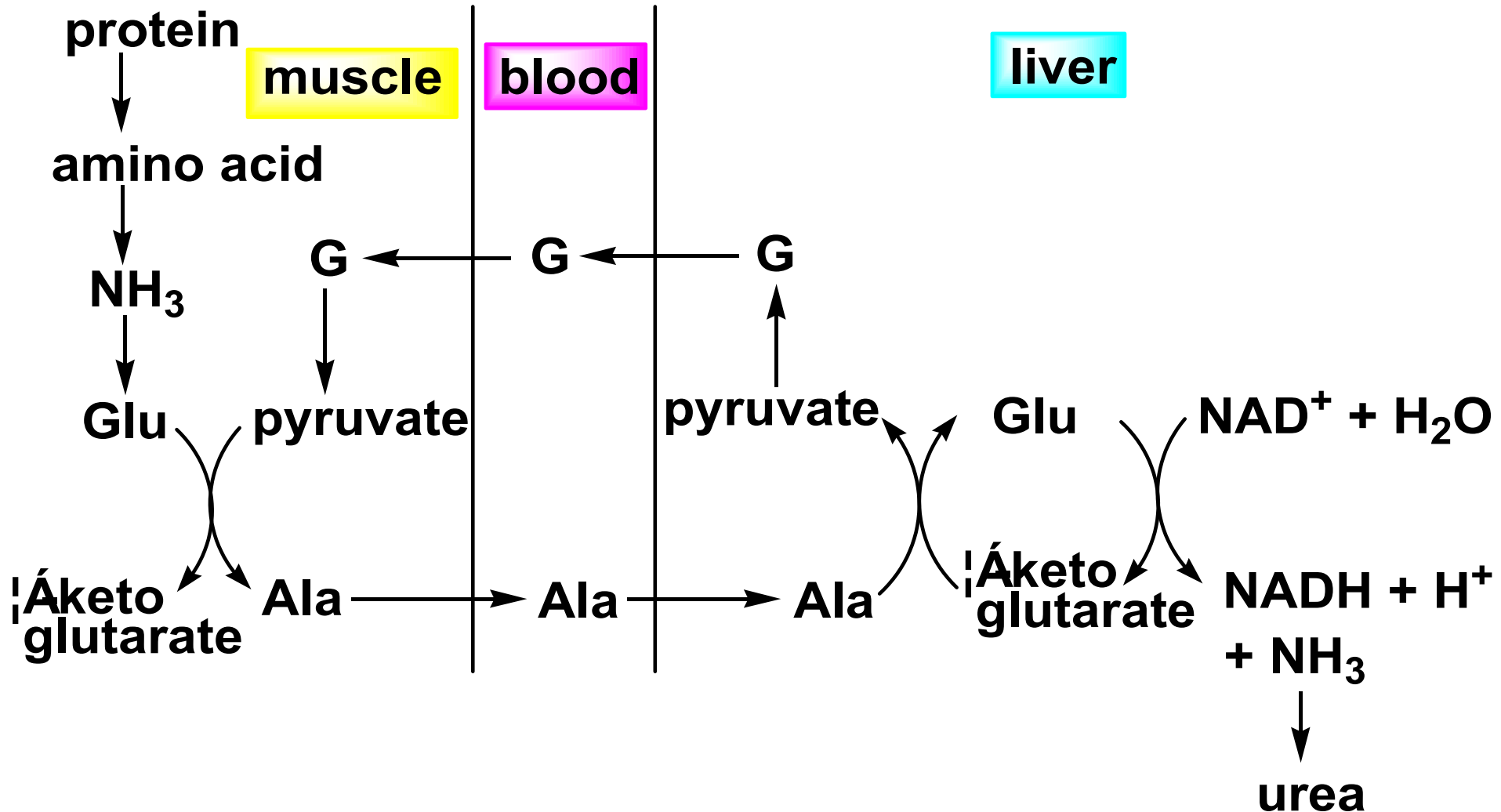
- (1) Formation of urea
- (2) Formation of Gln
- (3) Excrete in urine (NH_4^+)
- (4) Synthesis of AA

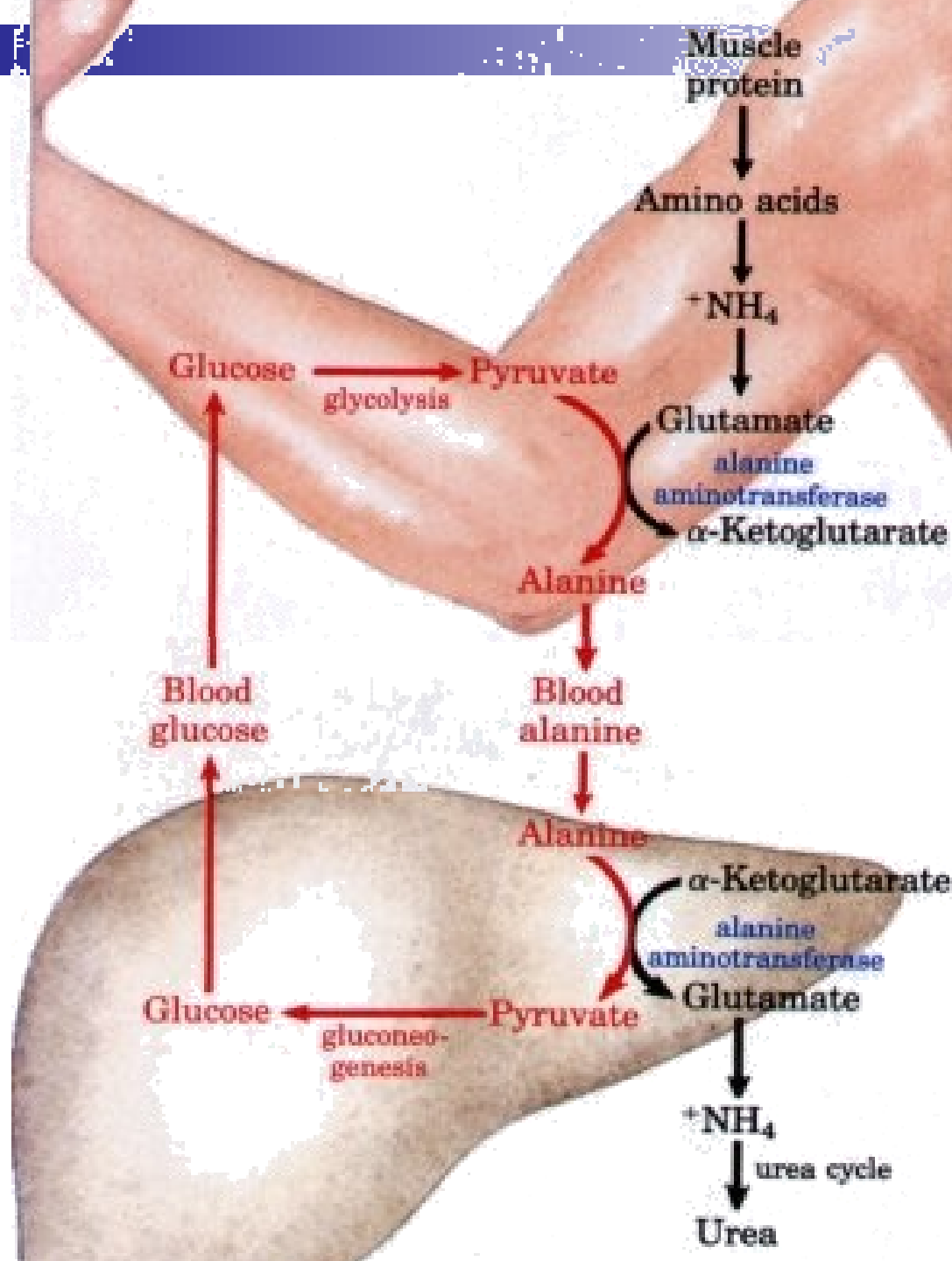


§ 4. 2 Transportation of NH_3

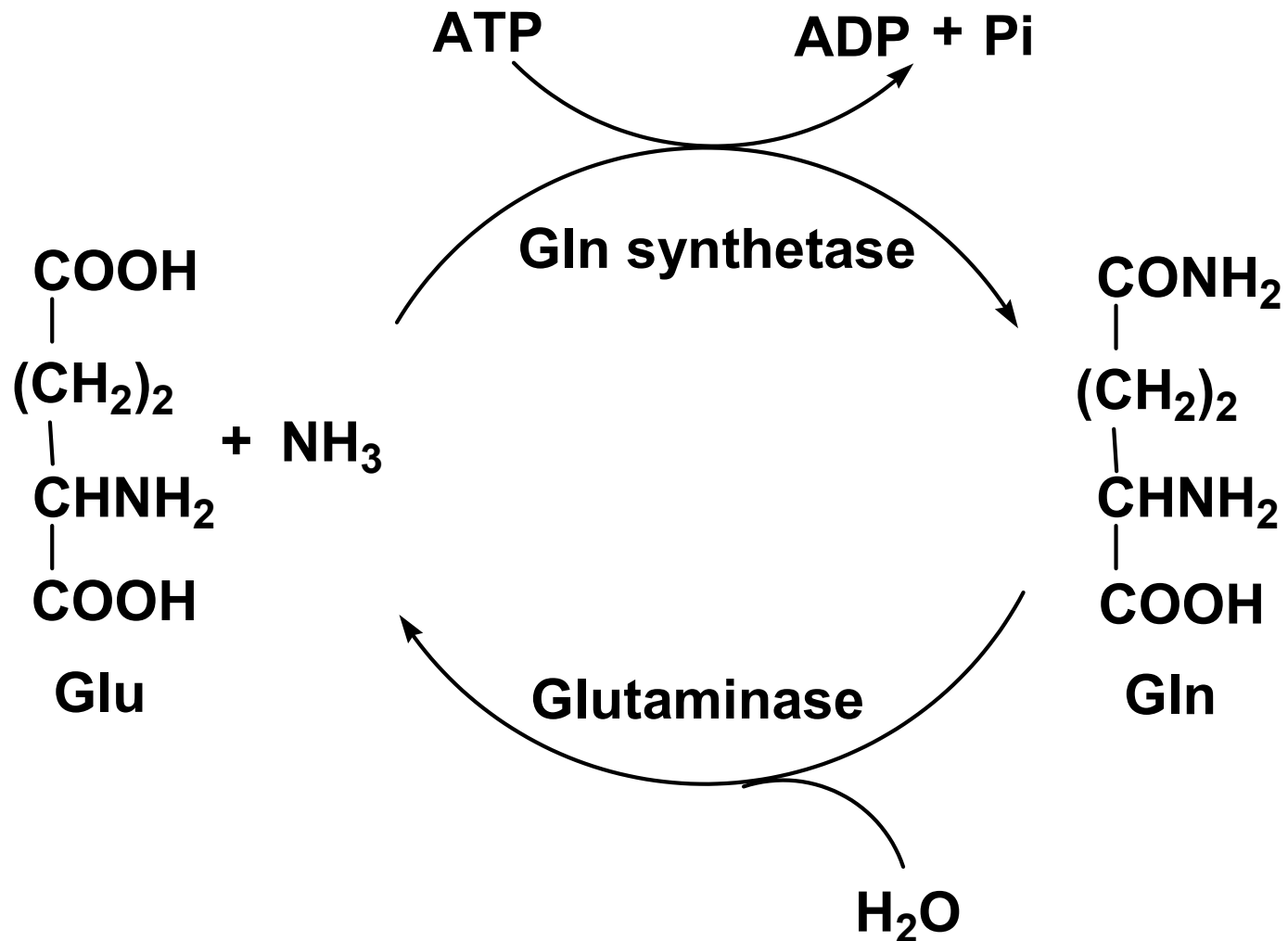
1. **Alanine**-glucose cycle
2. Transportation of ammonia by **Gln**

1. Alanine-glucose cycle

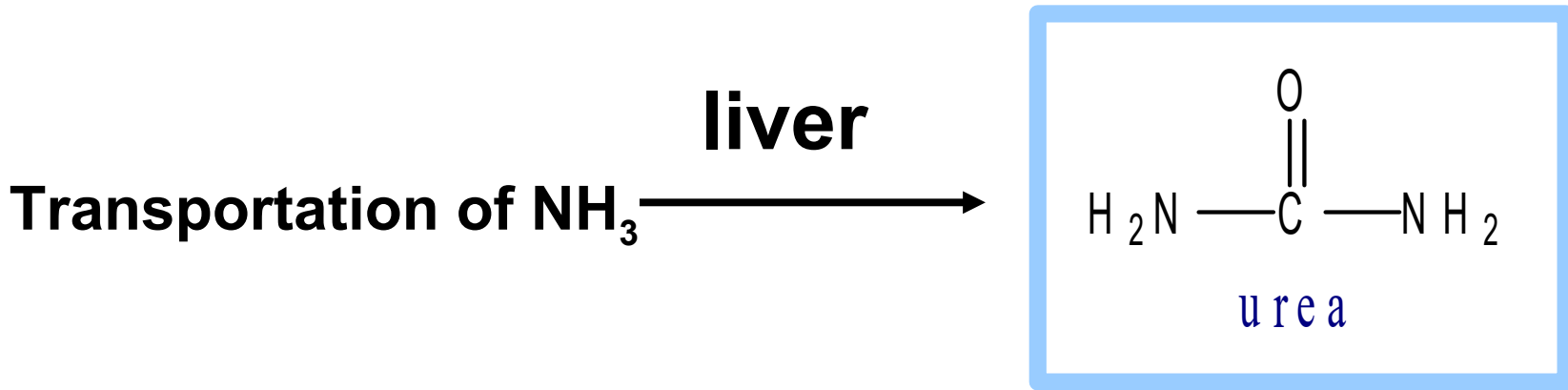




2. Transportation of ammonia by Gln



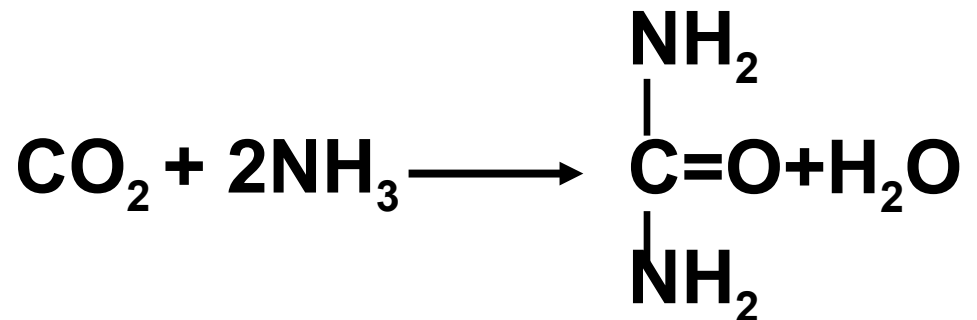
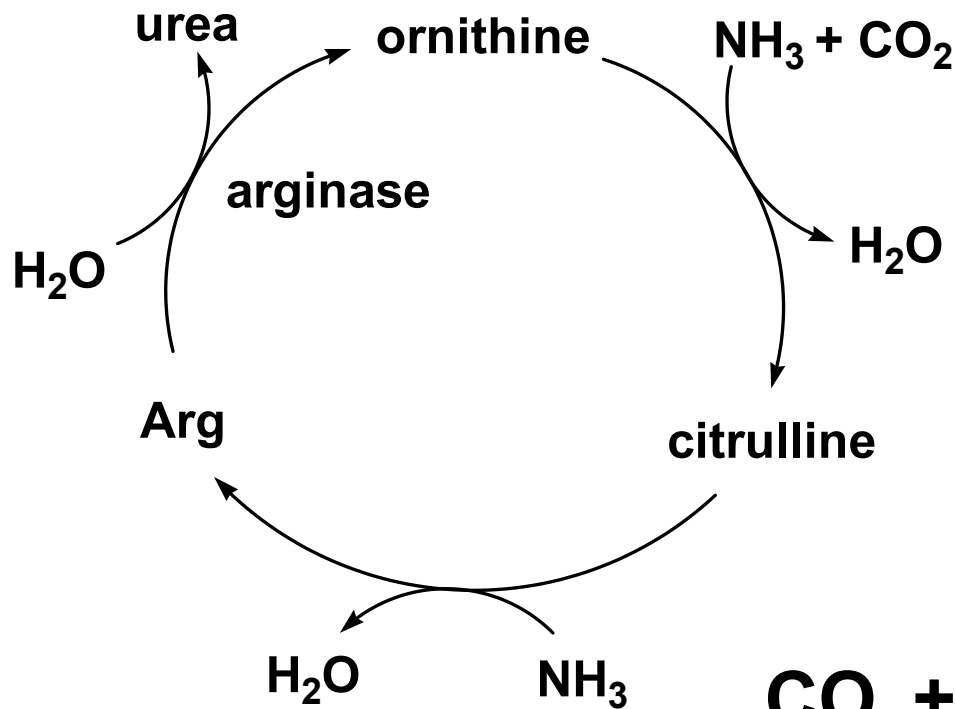
§ 4. 3 Formation of urea



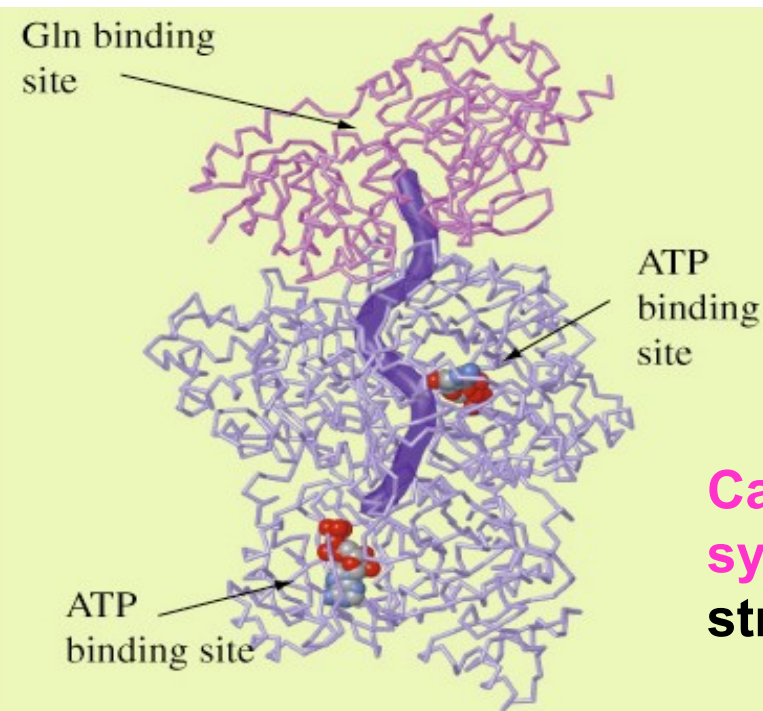
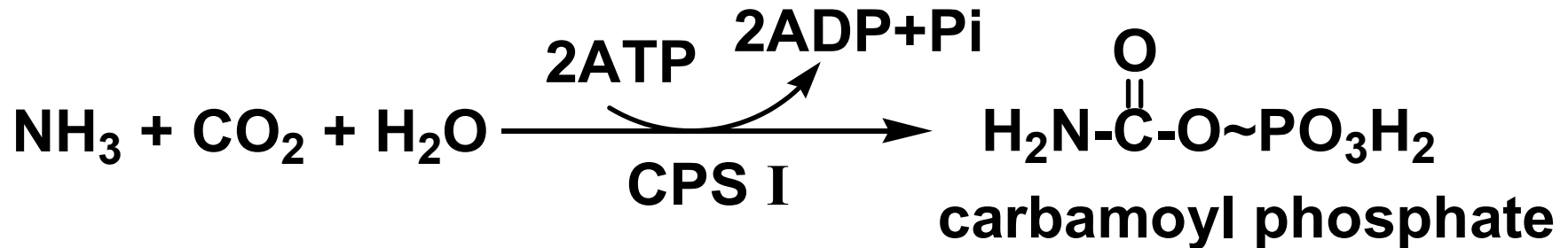
- ◆ Urea is **less toxic** than ammonia.
- ◆ The **Urea Cycle** occurs mainly in **liver**.
(**ornithine cycle / Krebs cycle**)
- ◆ Most animals convert excess nitrogen to **urea**, prior to excreting it.

1. Site: liver (mitochondria and cytosol)

2. Process ----- **Urea Cycle**



① Formation of carbamoyl phosphate (in mitochondria)



Carbamoyl phosphate synthase I

Carbamoyl phosphate synthase backbone structure

- Tunnel connecting active sites (blue wire)



Carbamoyl phosphate synthetase I :

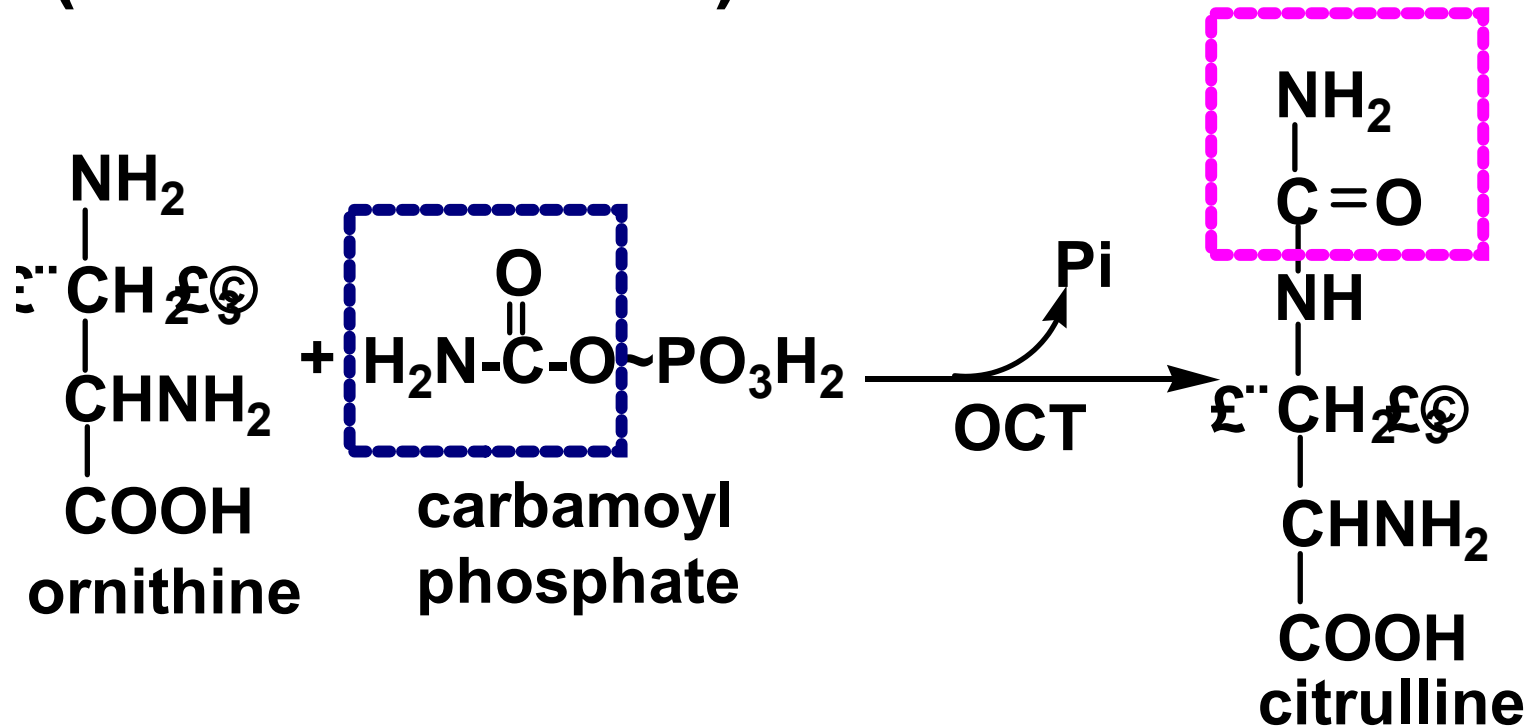
Occurs in mitochondria of liver cells. It is involved in urea synthesis.

Carbamoyl phosphate synthetase II:

Present in cytosol of liver cells which is involved in pyrimidine synthesis.

Carbamoyl phosphate synthetase I (CPS I) is an allosteric enzyme and is absolutely dependent up on **N-acetylglutamic acid (AGA)** for its activity.

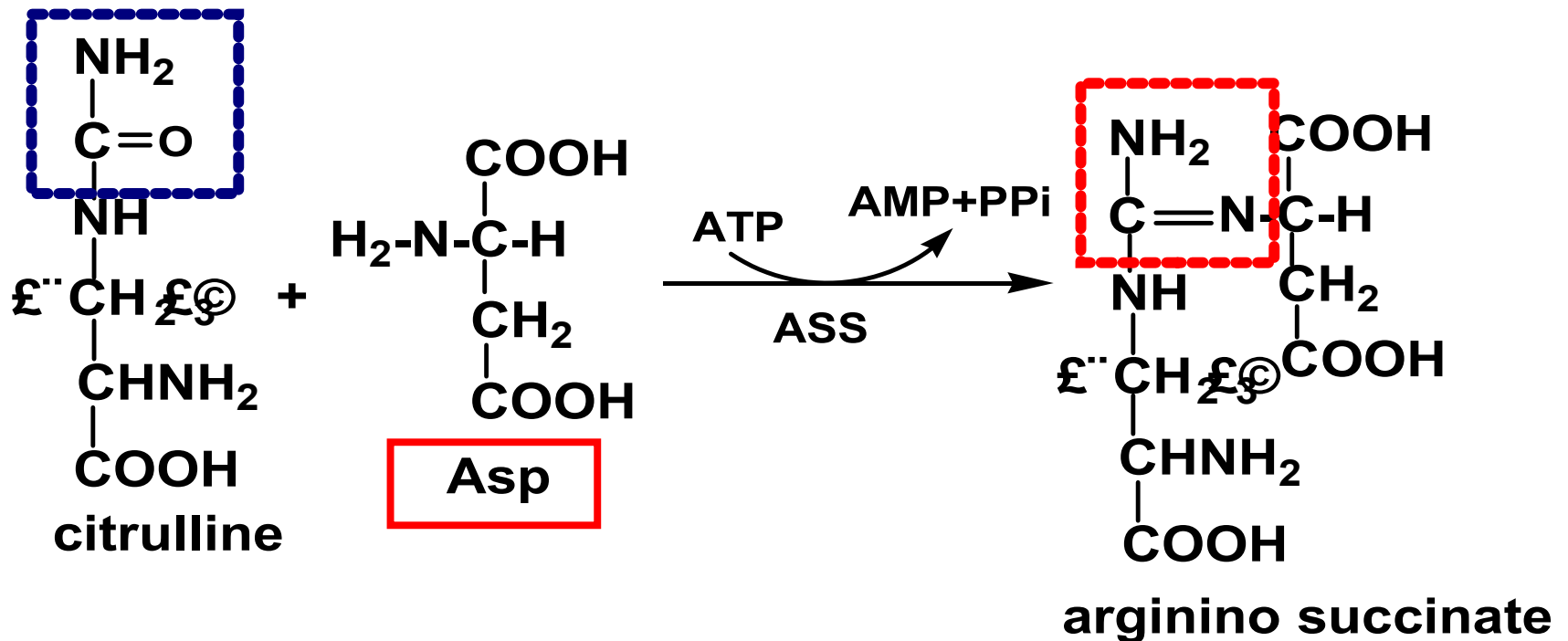
② Formation of citrulline (in mitochondria)



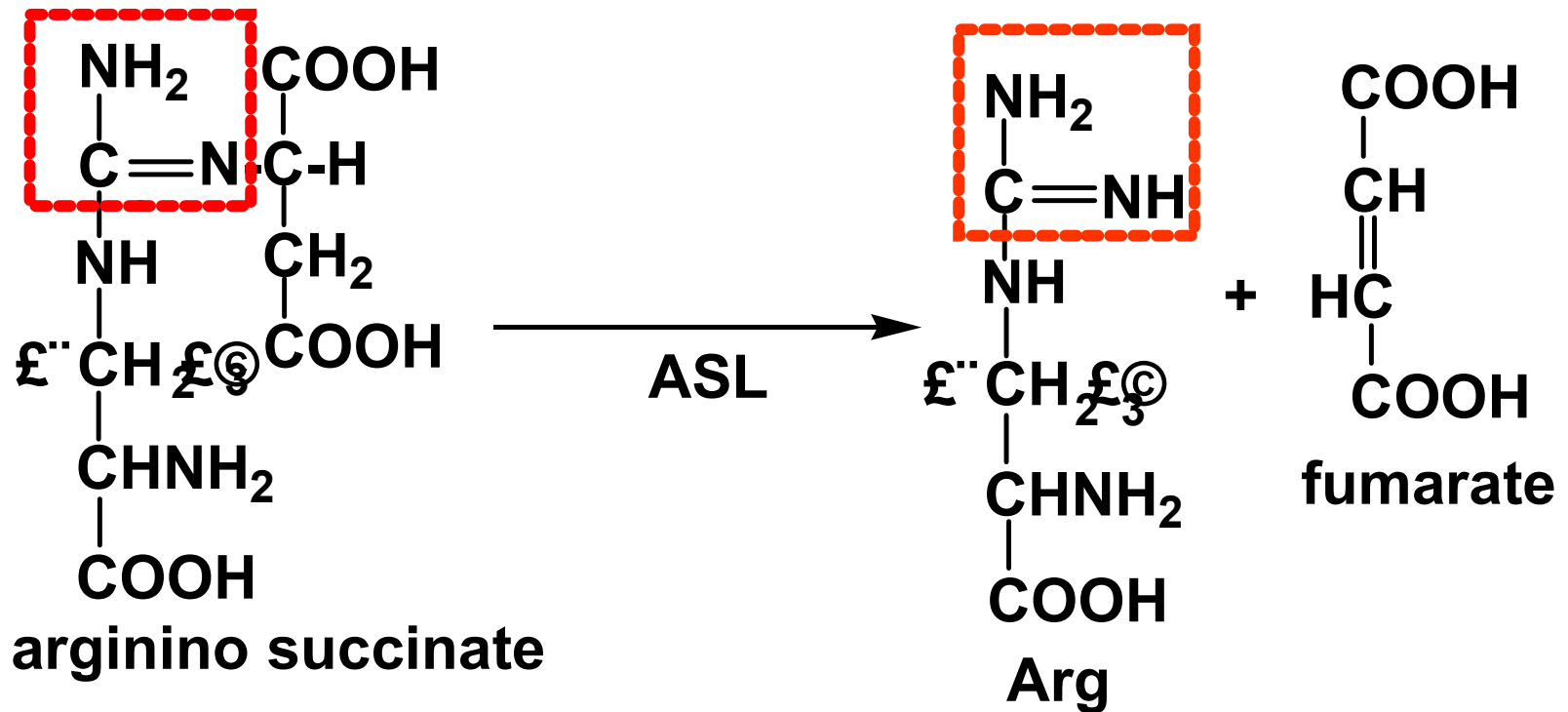
OCT: ornithine carbamoyl transferase

③ Formation of arginine (in cytosol)

two sub-steps

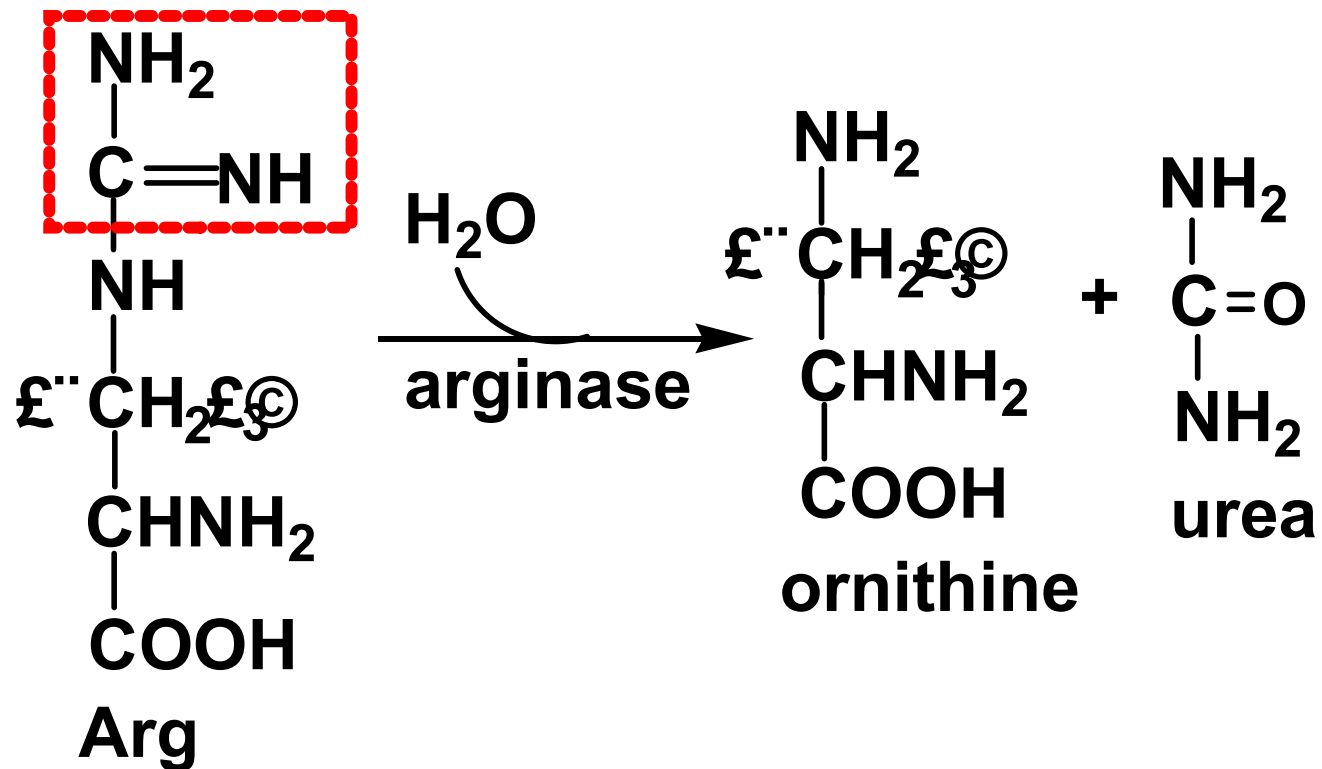


ASS: argininosuccinate synthetase

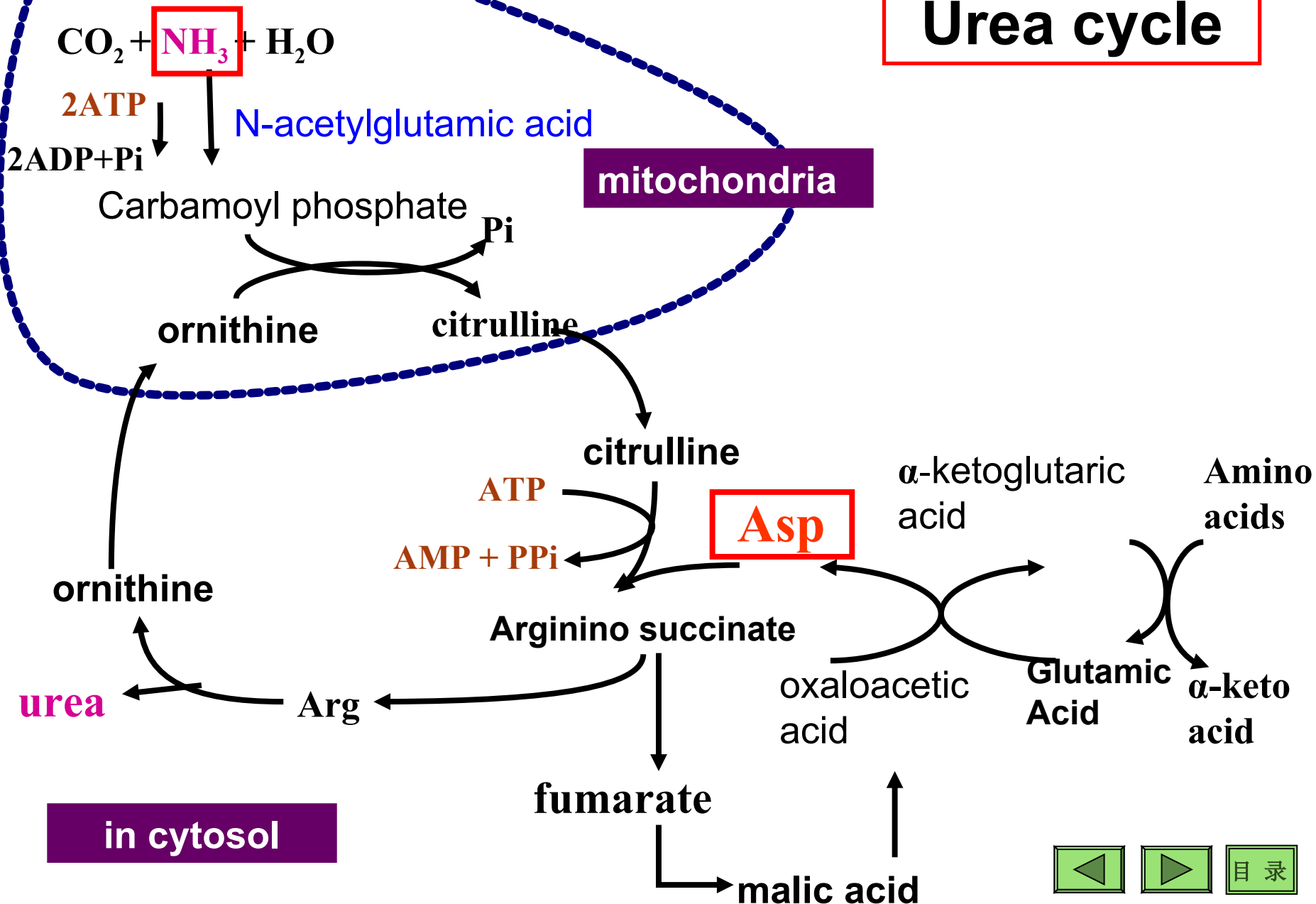


ASL: argininosuccinate lyase

④ Formation of urea (in cytosol)



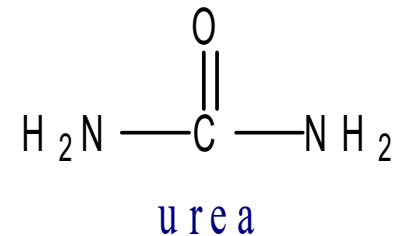
Urea cycle

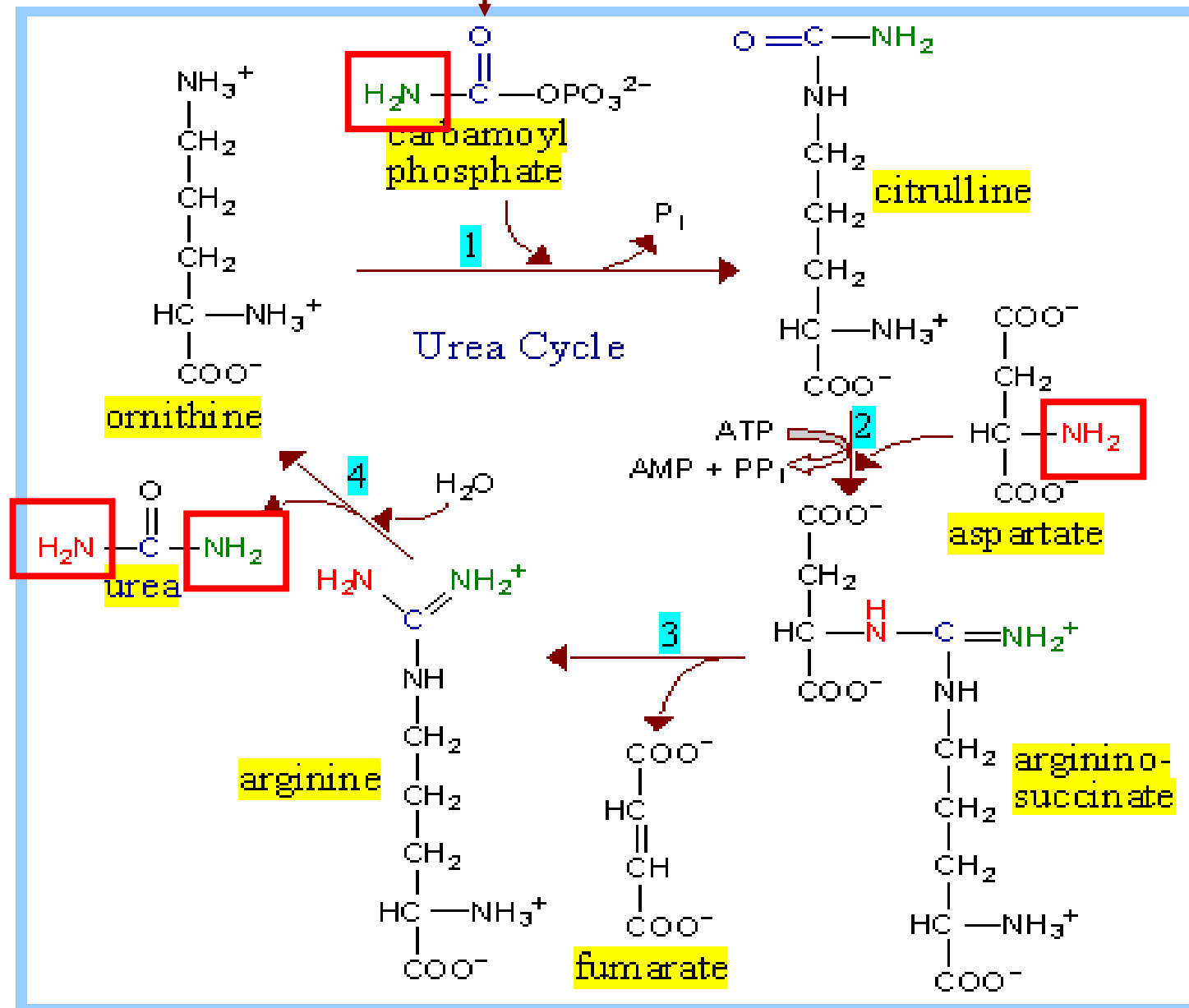


Summary of urea synthesis

Total formula :

- **One nitrogen** of urea molecule comes from ammonia, **another nitrogen** comes from Asp.
- **HCO₃⁻ ion** provides the carbon atom of urea.
- Found primarily in liver and lesser extent in kidney
- Synthesis of a urea will consume 3ATP and 4 ~P.





Regulation factors:

1. Ratio of protein in dietary foods:
2. Carbamoyl phosphate synthetase is allosterically activated by N-acetylglutamate
(acetyl CoA + glutamate $\rightarrow$ N-acetylglutamate)
3. Rate limiting enzyme: argininosuccinate synthetase(ASS)

Clinical significance of urea

A moderately active man consuming about 300gm carbohydrates, 100gm of fats and 100gm of proteins daily must excrete about 16.5gm of N daily.

95% is eliminated by the kidneys and the remaining 5%, for the most part as N, in the faeces.

Normal blood ammonia level:

in man, normal blood level of NH_3 varies from 40 to 70 $\mu\text{g}/100\text{ml}$. free NH_4^+ concentration of fresh plasma is less than 20 μg per 100ml.

HYPERAMMONEMIAS

Hyperammonemia is a metabolic disturbance characterised by **an excess of ammonia** in the blood. It is a dangerous condition that may lead to encephalopathy and death. It may be primary or secondary.

Ammonia has a direct **neurotoxic effect** on the CNS .for example ,elevated concentrations of ammonia in the blood cause the symptoms of ammonia intoxication, which include:

- tremors,
- slurring of speech,
- Somnolence ,vomiting ,cerebraledema
- ,
- and blurring of vision.



The two major types of hyperammonemia:

1. acquired hyperammonemia :

- ◆ dysfunction of liver is common cause of hyperammonemia(eg hepatic disease).
- ◆ porto-systemic encephalopathy: communications between portal and systemic veins.the portal blood may bypass the liver.

2. hereditary hyperammonemia:

is caused by several inborn errors of metabolism that are characterised by reduced activity of any of the enzymes in the urea cycle.



The major reasons of hyperammonemias:

1. Excessive putrefaction in the intestine,
example:hemorrhage of digestive tract.
2. Kidney secretion : kidney desfunction
Degradation of urea in the intestine
3. Liver desfunction or porto-systemic
encephalopathy,haemorrhage into GI tract.

hepatic encephalopathy

Hepatic encephalopathy is the occurrence of **confusion**, **altered level of consciousness** and **coma** as a result of excessive blood ammonia. it is also called **hepatic coma** or **coma hepaticum**. It may ultimately lead to death.

Postulated mechanisms for toxicity of high [ammonia]:

1. Depletion of glutamate & high ammonia level would drive Glutamate Dehydrogenase reaction to reverse:



The resulting depletion of α -ketoglutarate, an essential Krebs Cycle intermediate, could impair energy metabolism in the brain.

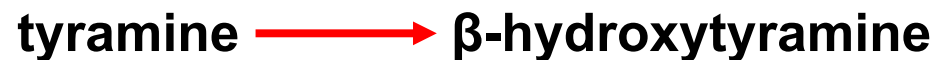
2. High [NH₃] would drive **Glutamine Synthase**:



This would deplete glutamate – a neurotransmitter & precursor for synthesis of the neurotransmitter GABA.

3. [glutamine][↑], cells swelling

4. false neurotransmitter:





Treatment of deficiency of Urea Cycle enzymes (some treatments depend on which enzyme is deficient):

- ◆ limiting protein intake to the amount barely adequate to supply amino acids for growth, while adding to the diet the α -keto acid analogs of essential amino acids.
- ◆ Liver transplantation has also been used, since liver is the organ that carries out Urea Cycle.



Thank You!!